# SOLID STATE

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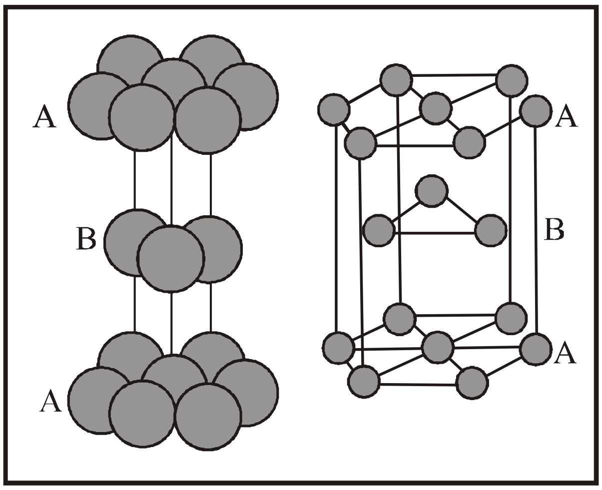


# THEORY

## 1. INTRODUCTION

Solid state is a state of matter besides liquid and gaseous state. In case of solids the inter molecular forces are very strong and empty spaces between the atoms/ions/ molecules is very less. That is why they have a fixed shape and volume.

### 1.1 Characteristic Properties of Solids

Solids are characterised by the following properties

* High density
* Low compressibility
* Rigidity
* Definite shape and volume.

## 2. CLASSIFICATION OF SOLIDS

Solids are broadly classified on the basis of following parameters.

* based on various properties
* based on bonding present in building blocks.

### 2.1 On the basis of various properties

Based on their various properties solids can be classified as

* Crystalline solids
* Amorphous solids.

Crystalline solids have a regular structure over the entire volume and sharp properties whereas amorphous solids have irregular structure over long distances and properties are not that sharp. Various differences are listed in table below

| Property | Crystalline solids | Amorphous solids |
|---|---|---|
| Shape | They have long range order | They have short range order. |
| Melting point | They have definite melting point | They do not have definite melting point. |
| Heat of fusion | They have a definite heat of fusion | They do not have definite heat of fusion |
| Compressibility | They are rigid and incompressible | These may be compressed to some extent |
| Cutting with a sharp edged tool | They are given cleavage i.e. they break into two pieces with plane surfaces | They give irregular cleavage i.e. they break into two pieces with irregular surface |
| Isotropy and Anisotropy | They are anisotropic | They are isotropic |
| Volume change | There is a sudden change in volume when they melt. | There is no sudden change in volume on melting. |
| Symmetry | They possess symmetry | They do not possess any symmetry. |
| Interfacial angles | They possess interfacial angles. | They do not possess interfacial angles. |

## 2.2 Based on bonding

There are various type of solids based on type of bonding present in their building blocks. Various types of solids along with their properties are given in the table below.

**The different properties of the four types of solids are listed in Different Types of Solids**

| Type of Solid | Constituent | Bonding | Examples | Physical | Electrical | Melting |
|---|---|---|---|---|---|---|
| (1) Molecular Solids | | | | | | |
| (i) Non polar | Molecules | Dispersion or London forces | Ar, $CCl_4$, $H_2$, $I_2$, $CO_2$ | Soft | Insulator | Very low |
| (ii) Polar | | Dipole-dipole Interactions | HCl, $SO_2$ | Soft | Insulator | Low |
| (iii) Hydrogen bonded | | Hydrogen bonding | $H_2O$ (ice) | Hard | Insulator | Low |
| (2) Ionic solids | Ions | Coulombic or electrostatic | NaCl, MgO, ZnS, $CaF_2$ | Hard but brittle | Insulators in solid state but conductors in molten state and in aqueous solutions | High |
| (3) Metallic solids | Positive ions in a sea of delocalised electrons | Metallic bonding | Fe, Cu, Ag, Mg | Hard but malleable and ductile | Conductors in solid state as well as in molten state | Fairly high |
| (4) Covalent or network solids | Atoms | Covalent bonding | $SiO_2$ (quartz) SiC, C (diamond) AlN, | Hard | Insulators | Very high |
| | | | $C_{(graphite)}$ | Soft | Conductor (exception) | |

## 3. STRUCTURE OF CRYSTALLINE SOLIDS

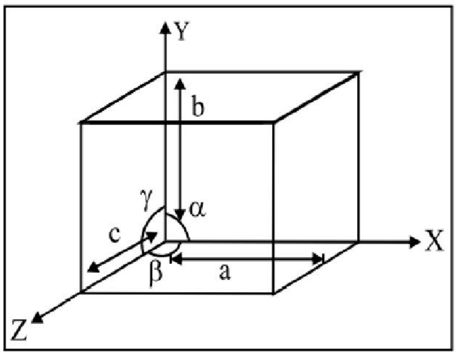
### 3.1 Crystal lattice and Unit Cell

The regular array of the building blocks (atoms/ions/molecules) inside the crystalline solid is called "Crystal Lattice".

The smallest part or crystal lattice which can be repeated in all directions to generate entire crystal lattice is called "Unit Cell".

In unit cell the atoms of ions or molecules are represented by small spheres. Various lattices are formed by variation in following parameters :

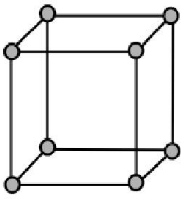
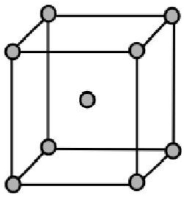
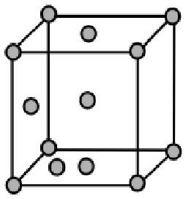
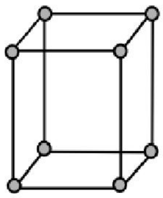
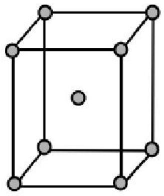
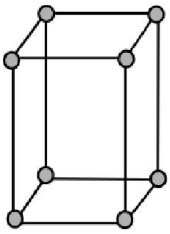
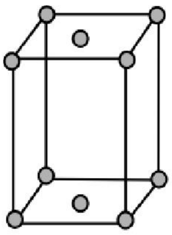
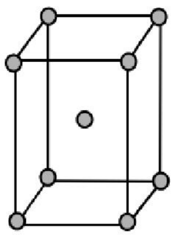
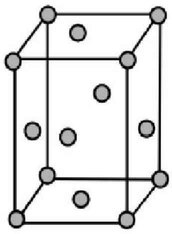
* The edge length along 3 axes - a, b, c
* The interfacial angles - $\alpha$, $\beta$, $\gamma$.
* Location of atom/ions w.r.t each other in crystal lattice.

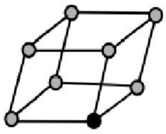
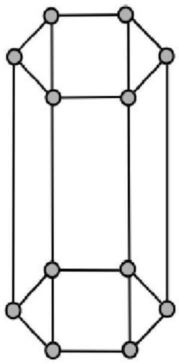
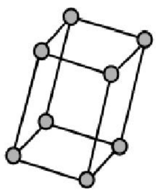
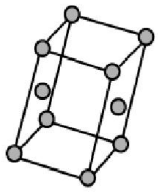
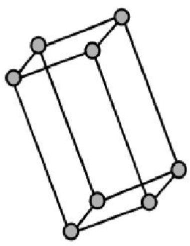



### 3.2 Primitive Unit Cells and Bravais Lattices

In all, there are seven types of unit cells and there can be some sub types of unit cells. These seven unit cells are called **Primitive Unit Cells** or **Crystal Habits.** Which are listed in table below

| Crystal System | Axial distances | Axial angles | Examples |
|---|---|---|---|
| Cubic | $a = b = c$ | $\alpha = \beta = \gamma = 90°$ | Copper, Zinc blende, KCl |
| Tetragonal | $a = b \neq c$ | $\alpha = \beta = \gamma = 90°$ | Sn (White tin), $SnO_2$, $TiO_2$ |
| Orthorhombic | $a \neq b \neq c$ | $\alpha = \beta = \gamma = 90°$ | Rhombic sulphur, $CaCO_3$ |
| Monoclinic | $a \neq b \neq c$ | $\alpha = \gamma = 90°; \beta \neq 90°$ | Monoclinic sulphur, $PbCrO_2$ |
| Hexagonal | $a = b \neq c$ | $\alpha = \beta = 90°; \gamma = 120°$ | Graphite, ZnO |
| Rhombohedral | $a = b = c$ | $\alpha = \beta = \gamma \neq 90°$ | $CaCO_3$ (Calcite), HgS (Cinnabar) |
| Triclinic | $a \neq b \neq c$ | $\alpha \neq \beta \neq \gamma \neq 90°$ | $K_2Cr_2O_7$, $CuSO_4.5H_2O$ |

| Crystal system | | Space lattice | | | Examples |
|---|---|---|---|---|---|
| **Cubic**<br>$a = b = c$<br>Here a, b and c are parameters (dimensions of a unit cell along three axes) size of crystals depend on parameters.<br>$\alpha = \beta = \gamma = 90°$<br>$\alpha$ $\beta$ and $\gamma$ are sizes of three angles between the axes. | **Simple :** Lattice Points at the eight corners of the unit cells.<br> | | **Body Centered :** Points at the eight corners and at the body centre.<br> | **Face Centered :** Points at the eight corners and at the six face centres.<br> | Pb, Hg, Ag, Au, Cu, ZnS, diamond, $KCl$, $NaCl$, $Cu_2O$, $CaF_2$ and alumns. etc. |
| **Tetragonal**<br>$a = b \neq c$,<br>$\alpha = \beta = \gamma = 90°$ | **Simple :** Points at the eight corners of the unit cell.<br> | | **Body Centered :** Points at the eight corners and at the body centre<br> | | $SnO_2$, $TiO_2$, $ZnO_2$, $NiSO_4$ $ZrSiO_4$, $PbWO_4$, white Sn etc. |
| **Orthorhombic**<br>(Rhombic)<br>$a \neq b \neq c$<br>$\alpha = \beta = \gamma = 90°$ | **Simple :** Points at the eight corners of the unit cell.<br> | **End Centered :** Also called side centered or base centered. Points at the eight corners and at two face centres opposite to each other.<br> | **Body Centered:** Points at the eight corners and and at the body centre<br> | **Face Centered :** Points at the eight coreners and at the six face centres.<br> | $KNO_3$, $K_2SO_4$, $PbCO_3$, $BaSO_4$ rhombic sulphur, $MgSO_4 . 7H_2O$ etc. |

| | | | |
|---|---|---|---|
| **Rhombohedral or Trigonal**<br>$a = b = c$,<br>$\alpha = \beta = \gamma \neq 90^\circ$ | **Simple :** Points at the eight corners of the unit cell<br> | | $NaNO_3$, $CaSO_4$<br>calcite, quartz<br>As, Sb, Bi etc. |
| **Hexagonal**<br>$a = b \neq c$,<br>$\alpha = \beta = 90^\circ$<br>$\gamma = 120^\circ$ | **Simple :** Points at the twelve or Points at the twelve<br>corners of the unit cell out corners of the haxagonal<br>lined by thick line. prism and at the centres of top and bottom faces.<br> | | ZnO, PbS, CdS,<br>HgS, graphite,<br>ice, Mg, Zn, Cd etc. |
| **Monoclinic**<br>$a \neq b \neq c$,<br>$\alpha = \gamma = 90^\circ$, $\beta \neq 90^\circ$ | **Simple :** Points ath the eight corners of the unit cell<br> | **End Centered :** Point at the eight corners and two face centres opposite to the each other.<br> | $Na_2SO_4.10H_2O$,<br>$Na_2B_4O_7.10H_2O$,<br>$CaSO_4.2H_2O$,<br>monoclinic sulphur etc. |
| **Triclinic**<br>$a \neq b \neq c$<br>$\alpha \neq \beta \neq \gamma \neq 90^\circ$ | **Simple :** Points at eight corners of the unit cell.<br> | | $CaSO_4$. $5H_2O$,<br>$K_2Cr_2O_7$, $H_3BO_3$ |

We will focus majorly on cubic unit cells and their arrangements.

## 3.3 Cubic Unit Cells

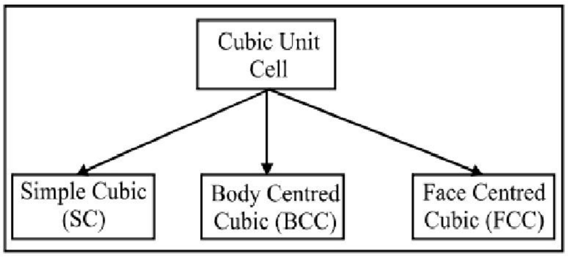
This is the most common unit cell. In a cubic unit cell there are following locations for the atoms or spheres

* Corners
* Body Centre
* Face Centres

Following are the contributions of a sphere kept at various locations.

| Location | Contribution |
|---|---|
| Corners | 1/8 |
| Body Centre | 1 |
| Face Centre | 1/2 |

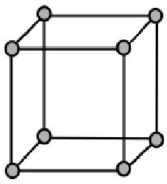
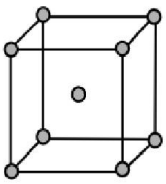
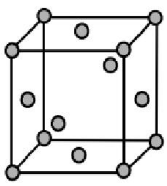
### 3.3.1 Types of cubic Unit Cells




These unit cells differ from each other in following factors.

* Location of spheres inside the unit cell.
* Rank of the unit cell (effective number of spheres inside a unit cell)
* Relation between edge length and radius of one sphere.
* Packing fraction (fraction of volume occupied by spheres in a unit cell)

The following parameters for all the 3 unit cells are listed is the table below :

| Type of Cublic Crystal | No. of atoms at different locations | | | Structure | Rank | Packing | Relation b/w atomic radius and edge length (a) |
|---|---|---|---|---|---|---|---|
| | Corners | Body centre | Face centre | | | | |
| Simple Cubic | 8 | – | – |  | 1 | 52% | r = a/2 |
| Body Centred | 8 | 1 | – |  | 2 | 68% | $r = \frac{\sqrt{3}a}{4}$ |
| Face Centred | 8 | – | 6 |  | 4 | 74% | $r = \frac{\sqrt{2}a}{4}$ |

### 3.4 Density of cubic crystals

Density of cubic crystal is given by the following formula

$$\text{Density } \rho = \frac{M \times Z}{a^3 \times N_A}$$

where, $Z \rightarrow$ rank of unit cell

$M \rightarrow$ Molar mass of solid

$a \rightarrow$ Edge length of unit cell

$N_A \rightarrow$ Avogadro Number.

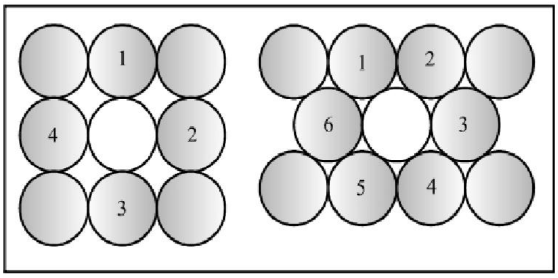
Valume of Z will depend upon the type of unit cell.

### 3.5 Close packing in Solids : Origin of unit cells

Suppose we have spheres of equal size and we have to arrange them in a single layer with the condition that spheres should come in close contact with each other. Two types of layers are possible :

1. **Square Packing**
2. **Hexagonal Packing**

In square packing spheres are placed in such a way that the rows have a horizontal as well as vertical arrangement. In this case Co-ordination Number is 4.




Hexagonal packing is more efficient. Its Co-ordination Number is 6 and voids in the packing are smaller than square packing.

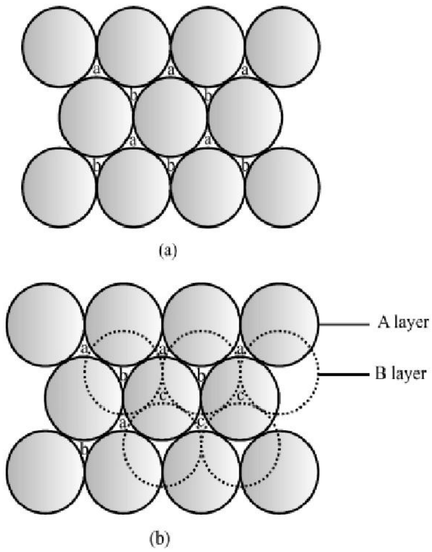
If we place another layer on square packing then there are following possibilities :

1. A similar layer placed just above foundation layer that is the spheres of the second layer coming just above the spheres of the first layer and layers get repeated. If first layer is termed A the packing in this case is AA... type and the unit cell is **simple cubic.**
2. On other hand if spheres of second layer are placed in depressions of first layer we get **BCC unit cell** and ABAB ..... type of packing.

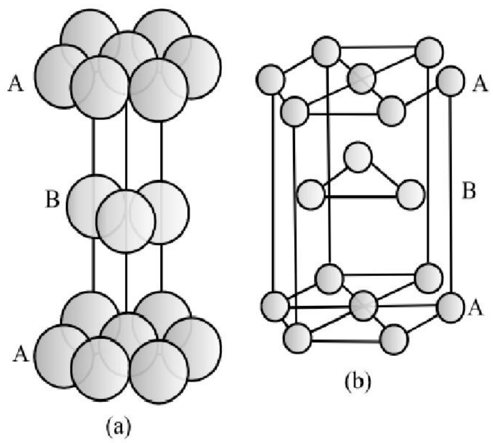
Arrangements based on hexagonal foundation layer are as follows :

If we put $2^{nd}$ layer in depressions of first hexagonal layer A two types of voids are created. X type of voids are those which are hollow and through voids of layer A and layer B. While Y type of voids are those voids of layer B which are exactly above spheres of layer A. If we place the spheres of 2nd layer on Y voids then we are repeating layer 1 and ABABAB.... type packing is obtained. In this arrangement hexagonal unit cell is obtained and packing is called **Hexagonal Close Packing** (HCP). The efficiency of this packing is 74%.

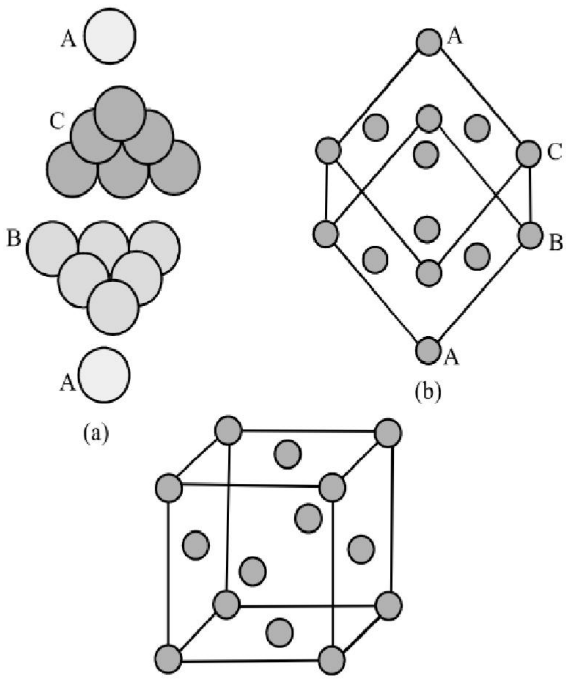
If the 3rd layer is placed on X-type of voids then a new layer C is obtained and then the arrangement will be repeated. We will obtain ABCABCABC..... type of packing. The unit cell for this arrangement is **FCC** and the packing effeciency is 74%.




**Packing of second layer (b) on first layer (a)**




**ABABA .... or hcp arrangement of spheres. Metals like magnesium, zinc, etc. adopt this type of arrangement.**




ccp arrangement
(c)

**ABCABCA ... or ccp arrangement of spheres**

## 4. VOIDS

### 4.1 Definition

The empty spaces inside a spheres are called "voids". The size and shape of voids depends upon the type of unit cell and packing.

### 4.2 Radius Ratio

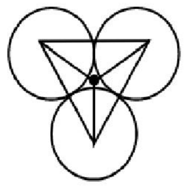
The size of void is expressed in terms of radius ratio of a sphere that can be exactly fit in the void to the radius of surrounding spheres. This expressed as :

Radius ratio $= \frac{r}{R}$

### 4.3 Types of voids

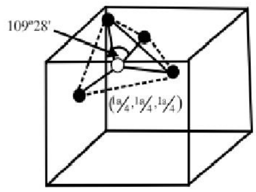
### 4.3.1 Trigonal void

It is a void formed of equal radii and touching each other as shown in figure.

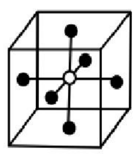
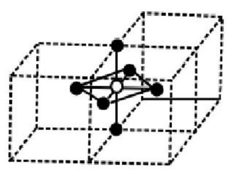
| Figure | Key Points |
|---|---|
|  Trigonal Void | * Radius Ratio<br>$\frac{r}{R} = 0.155$<br>* Smallest void<br>* Co-ordination no. = 3 |

### 4.3.2 Tetrahedral Void

It forms by contact of 4 spheres and is positioned at the centre of tetrahedron formed by contact of 4 spheres.

| Figure | Key Points |
|---|---|
|   An atom occupying a tetrahedral void | * Radius ratio $\frac{r}{R} = 0.225$<br>* No. of voids in FCC crystal = 8<br>* Position : at a distance<br>$\frac{a\sqrt{3}}{4}$ from every corner.<br>* Co-ordination no. = 4 |

### 4.3.3 Octahedral Void

| Figure | Key Points |
|---|---|
|   | * Radius ratio $\frac{r}{R} = 0.414$<br>* No. of voids in FCC crystal = 4<br>* Positions : Body centre<br>Edge centres<br>* Rank = 4<br>* Co-ordination No. = 6 |

### 4.3.4 Cubic Void

This void forms by close contact of 8 spheres

**Key Points**

* Radius Ratio = $\frac{r}{R} = 0.732$
* No. of voids in cubic crystal = 1
* Postion : at body centre
* Co-ordination number = 8
* Rank = 1

It is clear from above details that

Trigonal < Tetrahedral < Octahedral < Cubic
void void void void

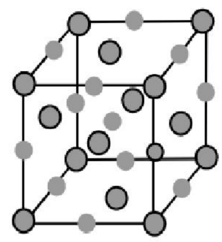
## 5. CLASSIFICATION OF IONIC STRUCTURES

Ionic compounds are formed by the simultaneous arrangement of cations and anions in lattice/unit cell. The larger of two species occupies major positions in a unit cell and the smaller ones occupy voids according to their size. Which is decided on the bases of radius ratio ($r_+/r_-$). The various ratios are listed below.

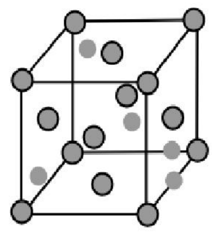
| Limiting Radius Ratios, | $x = r_+/r_-$ | C.N. | Shape Example |
|---|---|---|---|
| $x < 0.155$ | 2 | Linear | $BeF_2$ |
| $0.155 \le x < 0.225$ | 3 | Planar Triangular | $AlCl_3$ |
| $0.225 \le x < 0.414$ | 4 | Tetrahedron | ZnS |
| $0.414 \le x < 0.732$ | 6 | Octahedron | NaCl |
| $0.732 \le x < 0.999$ | 8 | Body Centred Cubic | CsCl |

Based on these ratio ranges, ionic crystal are classified into 5 categories which are as follows

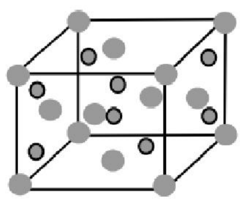
### 5.1 NaCl Type Structure

| Figure | Key Points |
|---|---|
|  Rock Salt Structure | * $Cl^-$ occupy corners and face centres and $Na^+$ occupy octhedral voids in FCC crystal.<br>* Effective formula = $Na_4Cl_4$<br>* Co-ordination No. of $Na^+$ = 6<br>* Co-ordination No of $Cl^-$ = 6<br>* Distance b/w nearest neighbours $[r_{Na^+} + r_{Cl^-} = \frac{a}{2}]$ |

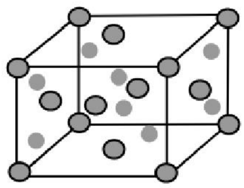
### 5.2 ZnS type Structure

| Figure | Key Points |
|---|---|
|  Zinc Blende Structure | * $S^{2-}$ ions occupy main positions and $Zn^{+2}$ ions are present in alternate tetrahedral voids in FCC crystal.<br>* Effective formula = $Zn_4S_4$<br>* Co-ordination of $Zn^{+2}$ = 4<br>* Co-ordination No. of $S^{2-}$ = 4<br>* $r_{Zn^{+2}} + r_{S^{2-}} = \frac{a\sqrt{3}}{4}$ |

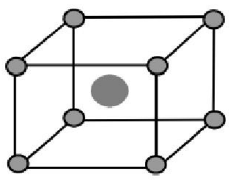
### 5.3 Fluorite Type Structure

| Figure | Key Points |
|---|---|
|  Fluorite Structure | * $Ca^{+2}$ ions occupy main positions and $F^-$ ions occupy tetrahedral voids in FCC crystal.<br>* Effective formula = $Ca_4F_8$<br>* Co-ordination no. of $Ca^{+2}$ = 8<br>* Co-ordination no. of $F^-$ = 4<br>* $r_{Zn^{+2}} + r_{S^{2-}} = \frac{a\sqrt{3}}{4}$ |

### 5.4 Anti Fluorite structure

| Figure | Key Points |
|---|---|
|  Anti-Fluorite Structure | * Common in alkali oxides like $Na_2O$, $Li_2O$ etc.<br>* $O^{2-}$ ions occupy FCC and $Li^+$ ions occupy the tetrahedral voids<br>* C.N. of $Li^+ = 4$<br>* CN of $O^{2-} = 8$<br>* $r_{Li^+} + r_{O^{2-}} = \frac{a\sqrt{3}}{4}$ |

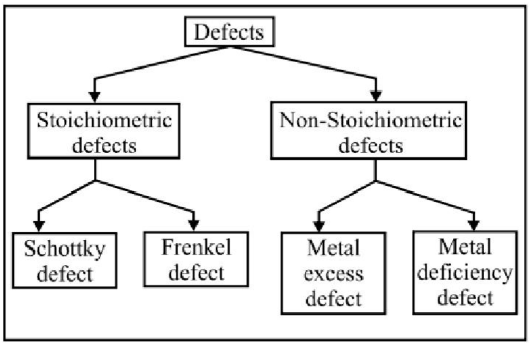
### 5.5 CsCl Type Structure

| Figure | Key Points |
|---|---|
|  Cesium Halide Structure | * $Cl^-$ ions simple cubic Locations (corners) and $Cs^+$ ions occupy body centre in BCC lattice<br>* Effective formula = CsCl<br>* Co ordination no of $Cs^+ = 8$<br>* Co ordination no. of $Cl^- = 8$<br>* $r_{Cs^+} + r_{Cl^-} = \frac{a\sqrt{3}}{2}$. |

## 6. IMPERFECTIONS IN SOLIDS

Sometimes some defects or imperfections occur in crystal structure.

### 6.1 Classification of defects

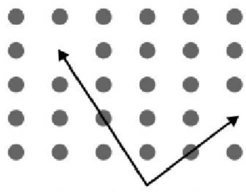


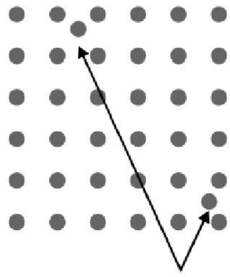

### 6.2 Vacancies

These are defects that occur when positions that should contain atoms or ions are vacant.

### 6.3 Interstitial sites

These are sites located between regular positions sometimes atoms or ions may occupy these positions.


(a) Lattice Vacancies

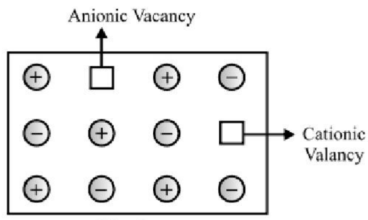

(b) Interstitial Atoms

### 6.4 Stoichiometric Defects

These defects do not disturb stoichiometry of solid substance.

### 6.4.1 Schottky defects

It is a vacancy defect in ionic solids. No. of missing cations and anions is equal so electrical neutrality is maintained. This defect decreases the density of the substance. The defect is shown by ionic substances in which cation and anion are of almost similar sizes. eq. KCl, NaCl, AgBr etc.

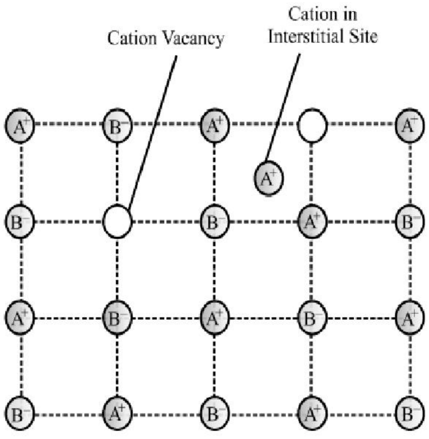


Schottky

### 6.4.2 Frenkel defect

In ionic solids the smaller ion is dislocated from its normal position to an interstitial site. It creates a vacancy defect at its original site and interstitial defect at new location. It is also called as dislocation defect. It does not change the density of solid.

This type of defect is shown by ionic substances in which there is a large difference in size of ions. eq ZnS, AgCl, AgBr etc.




A Frenkel defect

*Note..*

AgBr shows both Schottky and Frenkel defects.

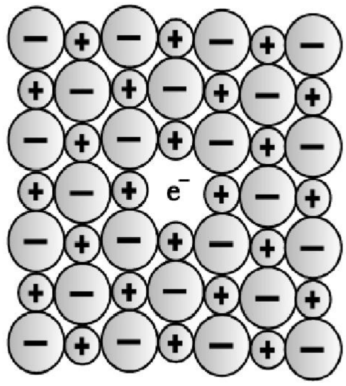
## 6.5 Non Stoichiometric Defects

The compounds having these defects contain combining elements in a ratio different from required by their stoichiometric formulae.

### 6.5.1 Metal Excess Defect

**Due to anionic vacancies :** The anion may be missing from its lattice site leaving an $e^-$ behind so that charge remains balanced. The site containing electron is called F centre. They import colour to the crystal, F stands for **Farbenzenter** meaning colour.

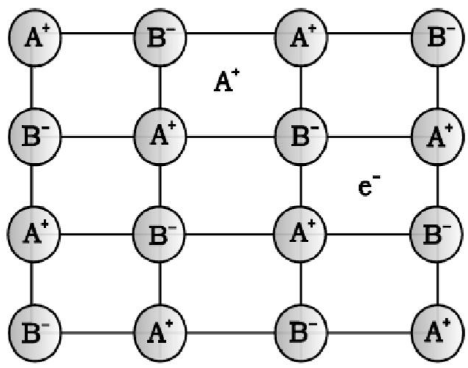
This defect is similar to schottky defect and is found is crystals having schottky defect eq. NaCl, KCl etc.



An F-centre in a crystal

**Due to the presence of extra cations in the interstitial sites.**

An extra cation may be present in interstitial site and an electron is present in another interstitial site so that electrical neutrality is maintained. This is similar to Frenkel defect and if found in crystal having Frenkel defect.



Metal excess defect caused by extra cation in the interstitial site.

### 6.5.2 Metal Deficiency Defect

This defect occurs when metal shows variable valency. eq. FeO is mostly found is varying compostions between $Fe_{0.93}O$ to $Fe_{0.96}O$. In crystals of FeO some $Fe^{+2}$ cations are missing and the loss of positive charge is made up by the presence of required number of $Fe^{+3}$ ions.

# SOLVED EXAMPLES

**Example - 1**

**Why are solids rigid ?**

**Sol.** Solids are rigid because the intermolecular forces of attraction that are present in solids are very strong. The constituent particles of solids cannot move from their positions, that is, they have fixed positions.However, they can oscillate about their mean positions.

**Example - 2**

**Why does urea have a sharp metling point but glass does not ?**

**Sol.** Urea is a crystalline solid, whereas glass is an amorphous solid. Crystalline solids have sharp melting points, whereas amorphous solids do not possess sharp melting points.

**Example - 3**

**Why does the window glass of the old buildings look milky?**

**Sol.** Due to continuous heating in the day and cooling at night, that is, annealing over a long period, glass acquires some crystalline characteristics. Glass is generally amorphous by nature, but due to annealing it becomes crystalline in nature. Hence, window glass of the old buildings looks milky.

**Example - 4**

**What difference in behavior between the glass and sodium chloride would you expect to observe if you break off a piece of either cube ?**

**Sol.** If we break or cut a piece of cube of sodium chloride, then we will observe that the surfaces on the two portions of the cube after breaking will be smooth but in case of glass crystals the two surfaces of the cube after breaking will be irregular. This is because of the cleavage property of crystalline solids according to which if we break or cut a crystalline solid then we get two smooth surfaces, but amorphous solids do not show cleavage property. Therefore in case of amorphous solids after cutting we get two irregular surfaces.

**Example - 5**

**What are crystalline solids ? State the categories of crystalline solids with examples.**

**Sol.** Crystalline solids are solids with highly ordered arrangement of constituent particles that is repeated many times throughout the structure. The pattern is so ordered and regular that arrangement of constituent particles at any site can be predicted, if arrangement at any one site is known. They are characterized by rigid structure, high and sharp melting points.

(a) Metallic solids (e.g., zinc, chromium, etc.)

(b) Ionic solids (e.g., sodium chloride, potassium bromide)

(c) Covalent solids (e.g., diamond, graphite)

(d) Molecular solids (e.g., iodine, naphthalene)

**Example - 6**

**What type of solids are electrical conductors, malleable and ductile.**

**Sol.** Metallic solids are electrical conductors, malleable and ductile.

**Example - 7**

**Identify the forces that must be overcome to cause melting in the following solids. Rank the compounds in the order of expected melting points (lowest one first) :**

**(a) Diamond ; (b) $CF_4$ ; $CrF_2$ ; (d) $SCl_2$.**

**Sol.** (a) Diamond Network (solid) covalent bonds

(b) $CF_4$ London dispersion forces

(c) $CrF_2$ Electrostatic forces

(d) $SCl_2$ London forces and dipole-dipole forces

**The order is :** $CF_4 < SCl_2 < CrF_2 <$ diamond.

**Example - 8**

**Compare and contrast the bonding in molecular solids and network covalent solids. Which would you expect to have a higher melting point, molecular solids or network covalent solids ?**

**Sol.** In molecular solids, the constituent particles are molecules. The forces operating between them are weak dispersion forces or dipole-dipole forces of attraction or hydrogen bonding depending on the type of molecular solid. In covalent or network solids, the constituent particles are non-metal atoms linked to adjacent atoms by covalent bonds throughout the solid.

Covalent or network solids have a higher melting point than molecular solids.

**Example - 9**

**Give the significance of a "lattice point".**

**Sol.** The significance of a lattice point is that each lattice point represents one constituent particle of the solid which may be an atom, a molecule (group of atoms), or an ion.

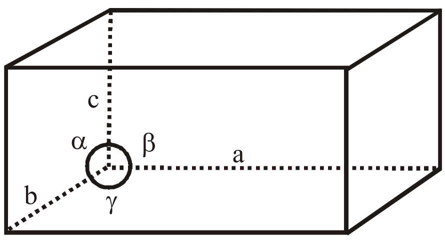
**Example - 10**

**Name the parameters that characterize a unit cell.**

**Sol.** The parameters that characterize a unit cell are as follows:

(a) Its dimensions along the three edges a, b and c. These edges may or may not be equal.

(b) The angles between the edges. These are represented by $\alpha$ (between edges b and c), $\beta$ (between edges a and c) and $\gamma$ (between edges a and b).




**Example - 11**

**Distinguish between**

**(a) Hexagonal and monoclinic unit cells.**

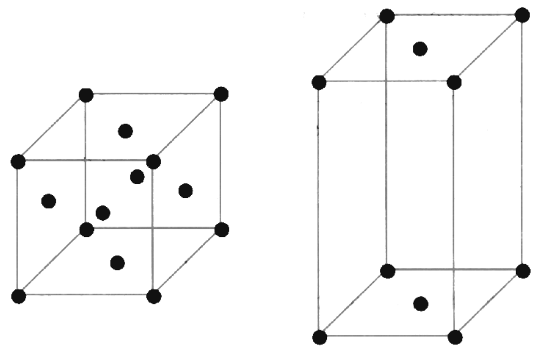
**(b) Face-centered and end-centered unit cells.**

**Sol.** (a) For a hexagonal unit cell, $a = b \neq c$ and $\alpha = \beta = 90^\circ, \gamma = 120^\circ$

For a monoclinic unit cell, $a \neq b \neq c$ and $\alpha = \gamma = 90^\circ, \beta = 90^\circ$

(b) A face-centered unit cell has one constituent particle present at the center of each face in addition to the particles present at the corners as shown in fig.

An end-centered unit cell has one constituent particle each at the center of any two opposite faces in addition to the particles present at the corners as shown in fig.



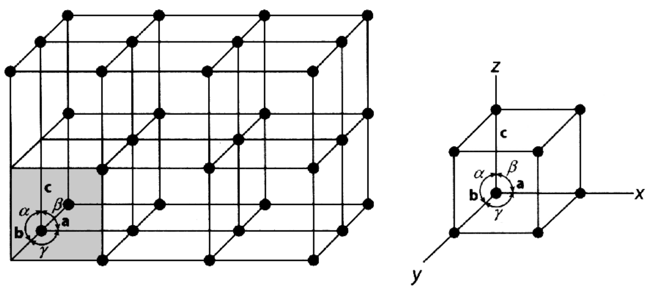
(a) (b)

**Example - 12**

**What relationship is there between a crystal lattice and a unit cell ?**

**Sol.** **Crystal lattice :** A regular arrangement of the constituent particles of a crystal in a three dimensional space is called crystal lattice or space lattice.

**Unit cell :** The smallest three-dimensional portion of a complete space lattice, which when repeated over and again in different directions produces the complete space lattice. The relation between unti cell and crystal lattice is that the unit cell is part of the lattice as shown in fig.




Space lattice and unit cell

**Example - 13**

**The unit cell of a substance has cations $A^+$ at the corners of the unit cell and the anions $B^-$ in the center. What is the simplest formula of the substance ?**

**Sol.** Number of cations ($A^+$) per unit cell = $8 \times 1/8 = 1$

Number of anions ($B^-$) per unit cell = 1

Hence, the ratio of $A^+ : B^-$ is 1 : 1 and the formula of the compound is AB.

**Example - 14**

**A solid substance AB has a rock salt geometry. What are the coordination numbers of A and B ? How many atoms of A and B are present in the unit cell ?**

**Sol.** The coordination number of A and B is 6. Four atoms each of A and B are present in the unti cell.

**Example - 15**

**MgO has the structure of NaCl, and TiCl has the structure of CsCl. What are the coordination numbers of the ions in MgO and TiCl ?**

**Sol.** The coordination number of ions in MgO is 6 and coordination number of ions in TiCl is 8.

**Example - 16**

**Explain how much portion of an atom is located at (a) corner and (b) body center of a cubic unit cell is part of its neighboring unit cell.**

**Sol.** (a) An atom located at the corner of a cubic unit cell is shared by eight adjacent unit cells. Therefore, 1/8th portion of the atoms belongs to one unit cell.

(b) An atom located at the body center of a cubic unit cell is not shared by any other unit cell. Hence, it belongs fully to the unit cell.

**Example - 17**

**How many copper atoms are there within the face-centered cubic unit cell of copper ?**

**Sol.** In the face-centered cubic unit cell, there are eight atoms at the corners and six atoms on the faces. Therefore, the contribution of atom at the corners $= 1/8 \times 8 = 1$, and contribution of atoms on the faces $= 1/2 \times 6 = 3$. Thus the number of copper atoms present is four per unit cell.

**Example - 18**

**A compound formed by elements A and B has a cubic structure in which A atoms are at the corners of the cube and B atoms are at the face centers. Derive the formula of the compound.**

**Sol.** As a atoms are present at the 8 corners of the cube, therefore number of atoms of A in the unit cell $= 1/8 \times 8 = 1$. As B atoms are present at the face centers of the cube, therefore number of atoms of B in the unit cell $= 1/2 \times 6 = 3$. Therefore, ratio of atoms A:B = 1:3. Hence, the formula of compound is $AB_3$.

**Example - 19**

**A cubic solid is made up of two elements X and Y. Atoms Y are present at the corners of the cube and atoms X at the body center. What is the formula of the compound ? What are the coordination numbers of X and Y ?**

**Sol.** Number of atoms of and number atoms of

Y = 1. The formula of the compound is XY.

The coordination number of the compound will be 8 because one X atom is surrounded by eight Y atoms at the corner of the cube.

**Example - 20**

**AgI crystallizes in cubic close-packed ZnS structure. What fraction of tetrahedral sites is occupied by $Ag^+$ ions ?**

**Sol.** In the face-centered unit cell of AgI, there are four $I^-$ ions. As there are four $I^-$ ions in the packing, therefore there are eight tetrahedral voids. Of these, half are occupied by silver cations.

**Example - 21**

**A compund form is formed by two elements M and N. The element N forms ccp and atoms of M occupy 1/3rd of tetrahedral voids. What is the formula of the compound ?**

**Sol.** The ccp lattice is formed by the atoms of the element N. Here, the number of tetrahedral voids generated is equal to twice the number of atoms of the element N. According to the questions, the atoms of element M occupy 1/3rd of the ratio of the number of atoms of M to that of N is M : N = (2/3) : 1 = 2:3. Thus, the formula of the compound is $M_2N_3$.

**Example - 22**

**Which of the following lattices has the highest packing efficiency : (a) Primitive cubic (b) body-centered cubic and (c) hexagonal close-packed lattice ?**

**Sol.** Hexagonal close-packed lattice has the highest packing efficiency of 74%. The packing efficiencies of primitive cubic and body-centered cubic lattices are 52.4% and 68%, respecitvely.

**Example - 23**

**A solid $A^+B^-$ has NaCl type close-packed structure. If the anion has a radius of 241.5 pm, what should be the ideal radius of the cation ? Can a cation $C^+$ having radius of 50 pm be fitted into the tetrahedral hole of the crystal $A^+B^-$ ?**

**Sol.** As $A^+B^-$ has NaCl structure, $A^+$ ions will be present in the octahedral voids. Ideal radius of the cation will be equal to the radius of the octahedral void because in that case, it will touch the anions and the arrangement will be close-packed. Hence,

$$\text{Radius of octahedral void } r_A^+ = 0.414 \times r_B^- = 0.414 \times 241.5\,\text{pm} = 100.0\,\text{pm}$$

$$\text{Radius of tetrahedral void} = 0.225 \times r_B^- = 0.225 \times 241.5\,\text{pm} = 54.3\,\text{pm}$$

As the radius of the cation $C^+$ (50 pm) is smaller than the size of the tetrahedral void, it can be placed into the tetrahedral void (but not exactly fitted into it).

**Example - 24**

**Barium crystallizes in a body-centered cubic structure in which the cell-edge length is 0.5025 nm. Calculate the shortest distance between neighboring barium atoms in the crystal.**

**Sol.** From the geometry of the body-centered cubic structure, we find the shortest distance between barium atoms will occur down the body diagonal where

$$\text{Diagonal} = \sqrt{3}a = 4r_{Ba}^+$$

$$r_{Ba}^+ = \frac{\sqrt{3}}{4} \times 0.5025 \times 10^{-7}\,\text{cm} = 2176 \times 10^{-8}\,\text{cm}$$

The shortest distance between barium atoms will be $2r_{Ba}^+$.

Thus, Ba-Ba distance = $4.352 \times 10^{-8}$ cm.

**Example - 25**

**A solid is made up of two elements P and Q. Atoms Q are in ccp arrangement while atoms P occupy all he tetrahedral sites. What is the formula of the compound ?**

**Sol.** Suppose number of atoms of element Q = N. So, numberof tetrahedral sites = 2N. Therefore, the number of atoms of element P = 2N. The ratio P : Q = 2N:N = 2:1. Hence, the formula of the compound is $P_2Q$.

**Example - 26**

**Predict the structure of MgO crystal and coordination number of its cation in which cation and anion radii are equal to 65 pm and 140 pm, respectively.**

**Sol.** Radius of the cation is $r^+$ = 65 pm and radius of anion is $r^-$ = 140 pm. Therefore, the radius ratio is

$$\frac{r^+}{r^-} = \frac{65\,\text{pm}}{140\,\text{pm}} = 0.464$$

Since the radius ratio (0.464) is in between 0.414 and 0.732, $Mg^{2+}$ ions occupy octahedral voids. Therefore, the coordination number of the cation is 6.

**Example - 27**

**The mineral haematite, $Fe_2O_3$ consists of a cubic close-packed array of oxide ions with $Fe^{3+}$ ions occupying interstitial positions. Predict whether the iron ions are in the octahedral holes. Radius of $Fe^{3+}$ = 65 pm and that of $O^{2-}$ = 145 pm.**

**Sol.** The radius ratio is

$$\frac{r^+}{r^-} = \frac{65\,\text{pm}}{145\,\text{pm}} = 0.45$$

This lies in the range 0.414 – 0.732. Hence, $Fe^{3+}$ ions will be in the octahedral holes.

**Example - 28**

**An element occurs in bcc structure with cell edge 300 pm. The density of the element is 5.2 g $cm^{-3}$. How many atoms of the element does 200 g of the element contain ?**

**Sol.** The edge length of unit cell a = 300 pm = $300 \times 10^{-10}$ cm, density of element $\rho$ = 5.2 $cm^{-3}$, mass of the element m = 200 g. Let the molecular mass of the element be M and number of moles be n then we have

$$n = \frac{m}{M} \Rightarrow M = \frac{m}{n}$$

For bcc structure, Z = 2. Now using the relation

$$\rho = \frac{ZM}{a^3 \times N_A} = \frac{Z\left(\frac{m}{n}\right)}{a^3 \times N_A} \Rightarrow n = \frac{Zm}{a^3 \times N_A \times \rho}$$

$$= \frac{2\times 200}{(300\times 10^{-10})^3 \times 6.02\times 10^{23}\times 5.2}$$

The required number of atoms will be

$$N = N_A \times n = 6.02 \times 10^{23} \times \frac{2\times 200}{(300\times 10^{-10})^3 \times 6.02\times 10^{23}\times 5.2}$$

$$= 2.85 \times 10^{24}$$

**Example - 29**

**The density of chromium metal is 7.2 g cm$^{-3}$. If the unit cell has edge length of 289 pm, determine the type of unit cell. (Atomic mass of Cr = 52 u; $N_A = 6.02 \times 10^{23}$).**

**Sol.** Edge length of unit cell $a = 289\text{ pm} = 289 \times 10^{-10}$ cm, molecular mass of chromium M = 52 g mol$^{-1}$, density of chromium metal $\rho$ = 7.2 g cm$^{-3}$. Let the number of atoms be Z. The density can be calculated using the relation

$$\rho = \frac{Z\times M}{a^3 N_A}$$

Therefore,

$$Z = \frac{\rho a^3 N_A}{M} = \frac{7.2\times(289\times 10^{-10})^3\times 6.023\times 10^{23}}{52} \approx 2$$

Since the unit cell has 2 atoms, hence it is body-centered cubic (bcc).

**Example - 30**

**An element with molar mass $2.7 \times 10^{-2}$ kg mol$^{-1}$ forms a cubic unit cell with edge length 405 pm. If its density is $2.7 \times 10^3$ kg m$^{-3}$, what is the nature of the cubic unit cell ?**

**Sol.** It is given that density of the element, $\rho = 2.7 \times 10^3$ ; molar mass, $M = 2.7 \times 10^{-2}$ kg mol$^{-1}$ ;

edge length, $a = 405\text{ pm} = 405 \times 10^{-12}\text{ m} = 4.05 \times 10^{-10}$ m. Applying the relation, for density, we get

(using $N_A = 6.023 \times 10^{23}$ mol$^{-1}$)

$$\rho = \frac{Z\times M}{a^3 \times N_A}$$

Therefore,

$$Z = \frac{\rho\times a^3 N_A}{M}$$

$$= \frac{2.7\times 10^3\,\text{kgm}^{-3}\times(4.05\times 10^{-10}\,\text{m})^3\times 6.022\times 10^{23}\,\text{mol}^{-1}}{2.7\times 10^{-2}\,\text{kg mol}^{-1}}$$

$$= 4.004 \approx 4$$

This implies that four atoms of the element are present per unit cell. Hence, the unit cell is face-centered cubic (fcc) or cubic close-packed (ccp).

**Example - 31**

**Gold (atomic radius = 0.144 nm) crystallizes in a face-centered unit cell. What is the length of a side of the cell ?**

**Sol.** For a face-centerd unit cell, $a = 2\sqrt{2}r$ It is given that the atomic radius, r = 0.144 nm. So,

$$a = 2\sqrt{2}\times 0.144 = 0.407\text{ nm}$$

Hence, length of a side of the cell = 0.407 nm.

**Example - 32**

**Silver crystallizes in a cubic close-packed structure. The radius of a silver atom is 1.44 Å. Calculate the density of Ag.**

**Sol.** The density is calculated as

$$\rho = \frac{Z\times M}{a^3\times N_A}$$

where for ccp structure, Z = 4 and r = 1.44 Å. The relationship between radius of an atom with the edge length a for ccp crystal structure is r = 0.3535a.

$$a = \frac{1.44}{0.3535} = 4.07\text{Å} = 4.07\times 10^{-8}\text{cm}$$

Substituting these values and using M for Ag = 107.87 g mol$^{-1}$, we get

$$\rho = \frac{4\times 107.87}{(4.07\times 10-8)^3\times 6.02\times 10^{23}} = 10.63\text{ g cm}^{-3}$$

**Example - 33**

**The atomic radius of nickel is 124 pm. Nickel crystallizes in face-centered cubic lattice. What is the length of the edge of unit cell expressed in pm and angstroms ?**

**Sol.** In a fcc, edge lenth = a and the radius of each atom be r. In this structure, atoms touch each other along the face-diagonal, there fore

Face diagonal = $\sqrt{2}a = 4r$

or $a = \frac{4r}{\sqrt{2}} = 2\sqrt{2} \times 124\,pm = 350.7\,pm$

thus, the unit cell edge length is 350.7 pm.

**Example - 34**

**A compound with fcc crystal structure has a density of $2.163 \times 10^3$ kg m$^{-3}$. Calculate the edge length of its unit cell. The molar mass of the compound is 58.2 g mol$^{-1}$**

**Sol.** For an element, the density of a crystal is given by

$$\rho = \frac{Z \times M}{a^3 \times N_A}$$

where Z is the number of particles present per unit cell; M is the atomic mass of the element ; a is the edge length of the unit cell; $N_A$ is the Avogadro constant ; $\rho$ is density of the crystal given as $2.163 \times 10^3$ kg m$^{-3}$. Since density is usually expressed in g cm$^{-3}$, so

$$\rho = \frac{2.163 \times 10^3 \times 10^3}{(1 \times 10^2)^3} = \frac{2.163 \times 10^6}{10^6} = 2.163\ g\,cm^{-3}$$

For fcc crystal structure, Z = 4 and we know that $N_A = 6.02 \times 10^{23}$ mol$^{-1}$ M = 58.2 g mol$^{-1}$. Rearranging Eq. (1) in terms of a, we get

$$a^3 = \frac{Z \times M}{\rho \times N_A} = \frac{4 \times 58.2\,g\,mol^{-1}}{2.163 \times 6.02 \times 10^{23} mol^{-1} g\,cm^{-3}}$$

$$= \frac{232.8\,g\,mol^{-1}}{13.02 \times 10^{23} g\,cm^{-3} mol^{-1}} = 17.880 \times 10^{-23} cm^3$$

$= 178.8 \times 10^{-24}$ cm$^3$

So, the edge length of the unit cell is

$a = (178.8)^{1/3} \times 10^{-8} = 5.634 \times 10^{-8}$ cm or 563.4 pm.

**Example - 35**

**A metal with atomic mass 27 has a density of 2.7 g cm$^{-3}$ and its unit cell has an edge of 4 Å, what is the nature of the crystal lattice for the metal ?**

**Sol.** The density is given by

$$\rho = \frac{Z \times M}{a^3 \times N_A}$$

Rearranging in terms of Z, and substituting values, we get

$$Z = \frac{\rho \times a^3 \times N_A}{M} = \frac{2.7 \times (4 \times 10^{-8})^3 \times 6.02 \times 10^{23}}{27}$$

$= 3.85 \approx 4$

which means that the nature of crystal lattice is face-centered cubic.

**Example - 36**

**A metal (atomic mass = 50) has a body-centered cubic crystal structure. The density of the metal is 5.96 g cm$^{-3}$. Find the volume of the unit cell ($N_A = 6.023 \times 10^{23}$ atoms mol$^{-1}$).**

**Sol.** Atomic mass of metal M = 50 g mol$^{-1}$ ; density of the metals is $\rho = 5.96$ g cm$^{-3}$. For body-centered cubic crystal, Z =2. Avogadro's cosntant $N_A = 6.023 \times 10^{23}$ atoms mol$^{-1}$. Let the volume of the crystal be V. Now, using relation for density

$$\rho = \frac{Z \times M}{V \times N_A} \Rightarrow V = \frac{Z \times M}{\rho \times N_A} = \frac{2 \times 50}{5.96 \times 6.023 \times 10^3}$$

$= 2.786 \times 10^{-23} cm^3$

Hence, volume of the unit cell = $2.786 \times 10^{-23}$ cm$^3$.

**Example - 37**

**Iron (II) oxide has a cubic structure and each unit cell has side 5 Å. If the density of the oxide is 4 g cm$^{-3}$, calculate the number of $Fe^{2+}$ and $O^{2-}$ ions present in each unit cell (molar mass of FeO = 72 g mol$^{-1}$, $N_A = 6.02 \times 10^{23}$ mol$^{-1}$.)**

**Sol.** Edge length of each unit cell of iron (II) oxide is a = 5 Å = $5 \times 10^{-8}$ cm. The density of iron (II) oxide is $\rho = 4$ g cm$^{-3}$; the molecular mass of iron (II) oxide is M = 72 g mol$^{-1}$ ; Avogadro's constant is $N_A = 6.02 \times 10^{23}$ mol$^{-1}$. Let the number of $Fe^{2+}$ and $O^{2-}$ ions present in each unit cell be Z. Now using the relation for density we get

$$\rho = \frac{Z \times M}{a^3 \times N_A} \Rightarrow Z = \frac{\rho \times a^3 \times N_A}{M}$$

$$= \frac{4 \times (5 \times 10^{-8})^3 \times 6.02 \times 10^{23}}{72} \approx 4$$

Hence the number of $Fe^{2+}$ and $O^{2-}$ ions present in each unit cell will be Z = 4.

**Example - 38**

**KBr has fcc structure. The density of KBr is 2.75 g cm$^{-3}$. Find the distance between $K^+$ and $Br^-$. (Atomic mass of Br = 80.0).**

**Sol.** For KBr which has fcc structure, Z = 4 ; Avogadro's constant $N_A = 6.02 \times 10^{23}$ mol$^{-1}$ ; density of KBr is $\rho = 2.75$ g cm$^{-3}$ ; the moelcular mass of KBr = 119 g mol$^{-1}$. Let the distance between $K^+$ and $Br^-$ be a. Now using the relation for density, we get

$$\rho = \frac{Z \times M}{a^3 \times N_A} \Rightarrow a^3 = \frac{Z \times M}{\rho \times N_A} = \left(\frac{4 \times 119}{2.75 \times 6.02 \times 10^{23}}\right)$$

$$a = \left(\frac{4 \times 119}{2.75 \times 6.02 \times 10^{23}}\right)^{1/3} \Rightarrow a = 6.6 \times 10^{-8} \text{ cm}$$

In an fcc, the ions/atoms touch each other along the face diagonal.

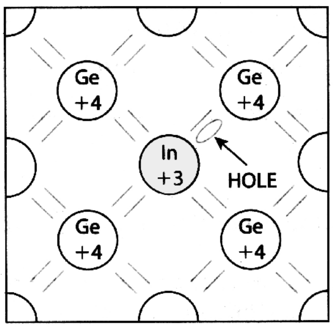
Face diagonal $= \sqrt{2}\ a = \sqrt{2} \times 6.6 \times 10^{-8}$ cm $= 9.334 \times 10^{-8}$ cm

The distance between $K^+$ and $Br^-$ = 1/2 (face dagonal) = $1/2 \times 9.33 \times 10^{-8}$ cm $= 4.667 \times 10^{-8}$ cm

**Example - 39**

**Classify each of the following as being either a p-type semiconductor : (a) Ge doped with In ; (b) S doped with B.**

**Sol.** (a) Ge (a Group 14 element) is doped In (a Group 13 element). We know that germanium is tetravalent, and that it is surounded by four other germanium atoms in close-packed structure. Now when an indium atom is added, which is trivalent, a hole is created due to the absence of an electron in lattice site as shown in fig. and the semiconductor generated will be a p-type semiconductor.




(b) When Si (a Group 14 element) is doped with B (a Grouop 15 element). We know that silicon is tetravalent by nature which means each silicon atom is surrounded by four other silicon atoms. When one boron atom (pentavalent in nature) is added to silicon atoms, then one extra electron is created in crystal lattice and the semiconductor generated will be called as n-type semiconductor.

**Example - 40**

**What type of defect can arise when a solid is heated ? Which physical property is affected by it and in what way ?**

**Sol.** When a solid is heated, vacancy defect can arise. A solid crystal is said to have vacancy defect when some of the lattice sites are vacant. Vacancy defect leads to a decrease in the density of the solid.

**Example - 41**

**What type of stoichiometric defect is shown by (a) ZnS and (b) AgBr ?**

**Sol.** (a) ZnS shows Frenkel defect.

(b) AgBr shows Frenkel as well as Schottky defects.

**Example - 42**

**Explain how vacancies are introduced in an ionic solid when a cation of higher valence is added as an impurity in it.**

**Sol.** When a cation of higher valence is added to an ionic solid as an impurity in it, the cation of higher valence repalces more than one cation of lower valence so as to keep the crystal electrically neutral. As a result, some sites become vacant. For example, when $Sr^{2+}$ is added to NaCl, each $Sr^{2+}$ ion replaces two $N^{a+}$ ions. However, one $Sr^{2+}$ ion occupies the site of one $Na^+$ ion and the other site remains vacant. Hence, vacancies are introduced.

**Example - 43**

**Ionic solids, which have anionic vacancies due to metal excess defect, develop color. Explain with the help of a suitable example.**

**Sol.** The color develops because of the presence of electrons in the anionic sites. These electrons absorb energy from the visible part of radiation and get excited. For example, when crystals of NaCl are heated in an atmosphere of sodium vapours, the sodium atoms get deposited on the surface of the crystal and the chloride ions leave their extra electrons at the lattice site and diffuse to the surface to form NaCl with the deposited Na atoms. The electron occupying the anionic lattice site is known as an F-center.

These electrons get excited by absorbing energy from the visible light and impart yellow color to the crystals.

**Example - 44**

**A Group 14 element is to be converted into n-type semiconductor by doping it with a suitable impurity. To which group should this impurity belong ?**

**Sol.** An n-type semiconductor conducts electricity because of the presence of extra electrons. Therefore, a Group 14 element can be converted to n-type semiconductor by doping it with a Group 15 element.

**Example - 45**

**What are the possible type of defects that can occur if a $Ca^{2+}$ ion replaces a $Na^+$ in a crystal lattice of NaCl ?**

**Sol.** When a $Ca^{2+}$ ion replaces a $Na^+$ in a crystal lattice of NaCl, one extra positive charge in introduced. The $Na^+$ vacancy or $Cl^-$ interstitial will be formed to balance the charge. Hence, the possible defects that can occur are Schottky defects.

**Example - 46**

**What are the two ways by which non-stoichiometric defects due to metal deficiency may occur ?**

**Sol.** Non-stoichiometry implies that either metal or non-metal atoms are present in excess. When non-metal atoms are in excess, two types of defects may arise due to metal deficiency. In the first type of defect, a positive ion is absent from its lattice site and the charge is balanced by adjacent metal (positive) ion having higher oxidation state, for example, FeO, FeS, etc. In the second type of defect, an extra negative ion is present in an interstitial position and the charge balance is maintained by the adjacent metal atom having higher charge.

**Example - 47**

**Which point defect in its crystal units alters the density of a solid ?**

**Sol.** Schottky defect decreases the density of the solid.

**Example - 48**

**What other element may be added to silicon to make electrons available for conduction of an electric current ?**

**Sol.** Silicon is used to create most semiconductors commercially. To make electrons available for conduction of an electric current, we add Group 15 elements (phosphorus, arsenic, antimony) to silicon which is tetravalent by nature. With the addition of each Group 15 element, an excess electron is created which results in conduction of the current.

**Example - 49**

**Why does Frenkel defect not change the density of AgCl crystals ?**

**Sol.** Frenkel defect does not change the density of AgCl crystals because in the Frenkel defect, the ions are not removed from the crystal. So there will be no change in the crystal structure, that is, there is no decrease in the number of ions. All the ions are inside the crystal, they are only dislocated.

**Example - 50**

**Mension one property which is caused due to the presence of F-center in a solid.**

**Sol.** One property which is caused due to the presence of F-center in a solid is appearance of color.

**Example - 51**

**Where are in the periodic table the atoms that are most likely to be involved in the formation of semiconductors located ?**

**Sol.** Semiconductors are typically formed by elements of Group 13–16, for example B, Si, Al, Ge.

**Example - 52**

**Why does germanium act as an n-type semiconductor ?**

**Sol.** Germanium acts as n-type semiconductor when a pentavalent impurity atom is added to pure germanium crystal. With addition of each impurity atom, one extra electron is created in the crystal lattice of germanium and hence the majority charge carriers are electrons in n-type semiconductors.

**Example - 53**

**What is the origin of the magnetic properties of an atom ?**

**Sol.** The origin of magnetic properties of an atom lies in the magnetic moment that arises from two types of motion of the electrons in the atom. An electron spins around its axis and moves in its orbital around the nucleus. These motions of the electron (charge particle) can be considered as a loop of current and impart permanent spin and an orbital magnetic moment to it. The magnitude of this magnetic moment is small and measured in the unit called Bohr magneton.

**Example - 54**

**What type of substances would make better permanent magnets, ferromagnetic or ferrimagnetic. Justify your answer.**

**Sol.** Ferromangetic substances would make better permanent magnets. In solid state, the metal ions of erromagnetic substansces are grouped together into small regions. These regions are called domains and each domain acts as a tiny magnet. In an unmagnified piece of a ferromagnetic substance, the domains are randomly oriented. As a result, the magnetic moments of the domain get cancelled. However, when the substance is placed in a magnetic field, all the domains get oriented in the direction of the magnetic field and a strong magnetic effect is produced. The ordering of the domains persists even after the removal of the magnetic field. Thus, the ferromagnetic substance becomes a permanent magnet.

**Example - 55**

**You are given marbles of diameter 10 mm. They are to be placed such that their centers are lying in a square bound byfour lines and each ot length 40 mm. What will be the arrangments of marbles in plane so that maximumnumber of marbles can be placed inside the area ? Sketch the diagram and derive expression for the number of molecules per unti area.**

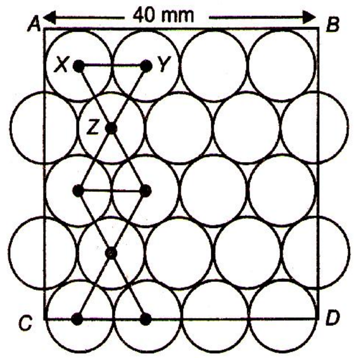
**Sol.** Maximum numberof marble balls or spheres can be accommodated if they are arranged in a hexagonal close packing (hcp).

Area of square having 4 lines each of 40 mm is

$= 40 \times 40 = 1600\ mm^2 = 16\ cm^2$

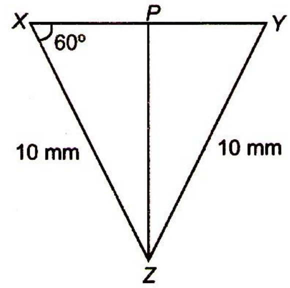
$(\because 40\ mm = 4\ cm)$

$= 4 \times 4\ cm^2$




Maximum no. of spheres = 14 (full) + 8 (half) = 18

Per unit area $= \frac{18}{16} = 1.125$



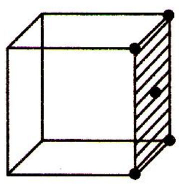

Length AB of square = 40 mm = 4 cm

Length AC = 4 cm.

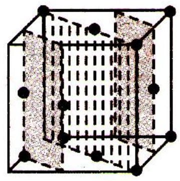
In $\Delta XYZ\ \frac{PZ}{XZ} = \sin 60^o = \frac{\sqrt{3}}{2}$

$PZ = 10\frac{\sqrt{3}}{2} = 5\sqrt{3}$

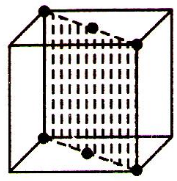
$AC = 5 + 4 \times 5\sqrt{3} = 5 + 20\sqrt{3} = 40\ mm = 4.0\ cm$



face-centred cubic



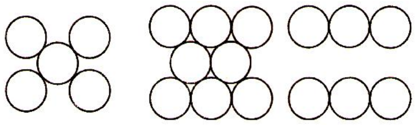
face-diagonal plane



diagonal plane

**Example - 56**

**The figures given below show the location of atoms in three crystallographic planes in a fcc lattice. Draw the unit cell for the corresponding structure and identify these planes in your diagorams.**



**Sol.** Face centred cube, face diagonal plane, diagonal plane.

**Example - 57**

**A unit cell of sodium chloride has four formula units. The edge length of the unit cell is 0.564 nm. What is the density of sodium chloride ?**

**Sol.** No. of atoms in fcc unit cell = 4

Mol. wt. of Nacl = 23 + 35.5 = 58.5

Length of edge of unit cell $= 0.564 \times 10^{-9}$ m

$= 0.564 \times 10^{-7}$ cm

Volume of unit cell $= (0.564 \times 10^{-7})^3$

$$\text{Density of NaCl} = \frac{4 \times 58.5}{(0.564 \times 10^{-7})^3 \times 6.023 \times 10^{23}}$$

$= 2.165$ g/cm$^3$

**Example - 58**

**The density of mercury is 13.6g/mL. Calculate approximately the diameter of an atom of mercury assuming that each atom is occupying a cube of edge length equal to the diameter of mercury atom.**

**Sol.** d = 13.6 g/mL

At. mass of Hg = 200 = M

1 g atomic wt. of Hg = 200 g = wt. of $6.023 \times 10^{21}$

No. of atoms present in 1 g of Hg

$$= \frac{6.023 \times 10^{23}}{200} = 3.0115 \times 10^{21}$$

Volume of 1 atom of Hg or unit cell

$$= \frac{1 \text{ g of Hg}}{\text{No. of atoms} \times \text{density}}$$

$$= \frac{1}{13.6 \times 3.0115 \times 10^{21}} = 2.44 \times 10^{-23} \text{ cm}^3$$

Edge length = Diameter of Hg atom

Diameter of 1 atom of Hg = (Volume of unit cell)$^{1/3}$

$= (2.44 \times 10^{-23})^{1/3}$

$d = (24.40 \times 10^{-24})^{1/3} = 2.91 \times 10^{-8}$ cm = 2.91 Å

# EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

## Characterstic Properties of Solid

**1.** Amorphous substances show

(1) Short and long range order

(2) Short range order

(3) Long range order

(4) Have no sharp M.P.

(a) 1 and 3 are correct (b) 2 and 3 are correct

(c) 3 and 4 are correct (d) 2 and 4 are correct

**2.** Amorphous solids are

(a) Solid substance in real sense

(b) Liquid in real sense

(c) Super cooled liquid

(d) Substance with definite melting point

**3.** Crystalline solids are

(a) Glass (b) Rubber

(c) Plastic (d) Sugar

**4.** A crystalline solid

(a) Changes abruptly from solid to liquid when heated

(b) Has no definite melting point

(c) Undergoes deformation of its geometry easily

(d) Has an irregular 3-dimensional arrangements

**5.** Which of the following is a molecular crystal

(a) SiC (b) Na*Cl*

(c) Graphite (d) Ice

**6.** Solid $CO_2$ is an example of

(a) Molecular crystal (b) Ionic crystal

(c) Covalent crystal (d) Metallic crystal

**7.** In graphite, carbon atoms are joined together due to

(a) Ionic bonding (b) Vander Waal's forces

(c) Metallic bonding (d) Covalent bonding

**8.** Mostly crystals show good cleavage because their atoms, ions or molecules are

(a) Weakly bonded together

(b) Strongly bonded together

(c) Spherically symmetrical

(d) Arranged in planes

## Crystal Lattice and Unit Cell

**9.** The number of spheres contained (i) in one body centred cubic unit cell and (ii) in one face centred cubic unit cell, is

(a) In (i) 2 and in (ii) 4 (b) In (i) 3 and in (ii) 2

(c) In (i) 4 and in (ii) 2 (d) In (i) 2 and in (ii) 3

**10.** The number of atoms or molecules contained in one body centered cubic unit cell is

(a) 1 (b) 2

(c) 4 (d) 6

**11.** If the number of atoms per unit in a crystal is 2, the structure of crystal is

(a) Octahedral (b) Body centred cubic

(c) Face centred cubic (d) Simple cubic

**12.** In fcc lattice, the neighbouring number of atoms for any lattice point is

(a) 6 (b) 8

(c) 12 (d) 14

**13.** In a face centred cubic arrangement of A and B atoms when A atoms are at the corner of the unit cell and B atoms of the face centres. One of the A atom is missing from one corner in unit cell. The simplest formula of compound is

(a) $A_7B_3$ (b) $AB_3$

(c) $A_7B_{24}$ (d) $A_{7/8}B_3$

**14.** An alloy of Cu, Ag and Au is found to have copper constituting the ccp lattice. If silver atoms occupy the edge centre and gold is present at body centre, the alloy has a formula

(a) $Cu_4Ag_2Au$ (b) $Cu_4Ag_2Au$

(c) $Cu_4Ag_3Au$ (d) CuAgAu

**15.** Sodium metal crystallizes as a body centred cubic lattice with the cell edge 4.29Å. What is the radius of sodium atom.

(a) $1.857 \times 10^{-8}$ cm (b) $2.371 \times 10^{-7}$ cm

(c) $3.817 \times 10^{-8}$ cm (d) $9.312 \times 10^{-7}$ cm

16. In a cubic structure of compound which is made from X and Y, where X atoms are at the corners of the cube and Y at the face centres of the cube. The molecular formula of the compound is

(a) $X_2Y$ (b) $X_3Y$

(c) $XY_2$ (d) $XY_3$

17. A compound is formed by elements A and B. This crystallizes in the cubic structure when atoms A are the corners of the cube and atoms B are at the centre of the body. The simplest formula of the compounds is

(a) AB (b) $AB_2$

(c) $A_2B$ (d) $AB_2$

## Density

18. Potassium has a *bcc* structure with nearest neighbour distance 4.52 Å. Its atomic weight is 39. Its density will be

(a) 454 kg $m^{-3}$ (b) 804 kg $m^{-3}$

(c) 852 kg $m^{-3}$ (d) 908 $kgm^{-3}$

19. The number of atoms in 100 g of an FCC crystal with density d = 10 g $cm^{-3}$ and cell edge as 200 pm is equal to

(a) $3 \times 10^{25}$ (b) $0.5 \times 10^{25}$

(c) $1 \times 10^{25}$ (d) $2 \times 10^{25}$

20. An element (atomic mass = 100 g/mol) having bcc structure has unit cell edge 400 pm. Then density of the element is

(a) 10.376 g/$cm^3$ (b) 5.188 g/$cm^3$

(c) 7.289 g/$cm^3$ (d) 2.144 g/$cm^3$

21. A crystal plane makes intercepts 2a, 2b and 3c on the crystallographic axes where a, b, c represent the intercepts of the unit plane. The Miller indices of the plane will be

(a) 223 (b) 332

(c) $\frac{1}{3}\frac{1}{3}\frac{1}{2}$ (d) $\frac{1}{2}\frac{1}{2}\frac{1}{3}$

22. The unit cell of aluminum is a cube with an edge length of 405 pm. The density of aluminium is 2.70 g $cm^{-3}$. What is the structure of unit cell of aluminium?

(a) Body–centred cubic cell

(b) Face-centred cubic cell

(c) End-centred cubic cell

(d) Simple cubic cell

23. How many lithium atoms are present in a unit cell with edge length 3.5 Å and density 0.53 $gcm^{-3}$? (Atomic mass of Li = 6.94)

(a) 2 (b) 1

(c) 4 (d) 6

24. An element with atomic mass 100 has a bcc structrue and edge length 400 pm. The density of element is

(a) 10.37 $gcm^{-3}$ (b) 5.19 g $cm^{-3}$

(c) 7.29 g $cm^{-3}$ (d) 2.14 g $cm^{-3}$

## Packing and Voids

25. A metal crystallises into a lattice containing a sequence of layer as AB AB AB ....... . What percentage of voids are left in the lattice?

(a) 72% (b) 48%

(c) 26% (d) 32%

26. The two dimensional coordination number of a molecule in square close packed layer is

(a) 4 (b) 2

(c) 3 (d) 6

27. Which of the following statements is not true about the voids?

(a) Octahedral void is formed at the centre of six spheres which lie at the apices of a regular octahedron.

(b) There is one octahedral site for each sphere.

(c) There are two tetrahedral sites for each sphere.

(d) Octahedral voids are formed when the triangular voids in second layer exactly overlap with similar voids in the first layer.

28. In ABC ABC packing if the numberof atoms in the unit cell is n then the number of tetrahedral voids in the unit cell is equal to

(a) n (b) n/2

(c) n/4 (d) 2n

29. In ccp arrangement the pattern of successive layers can be designated as

(a) AB AB AB (b) ABC ABC ABC

(c) AB ABC AB (d) ABA ABA ABA

## Ionic Structures

**30.** In which compound 4 : 4 coordination is found

(a) ZnS (b) CuCl

(c) AgI (d) All of these

**31.** The unit cell of a NaCl lattice

(a) Is body centred cube (b) Has 3$Na^+$ ions

(c) Has 4NaCl units (d) Is electrically charged

**32.** In CsCl structure, the coordination number of $Cs^+$ is

(a) Equal to that of $Cl^-$, that is 6

(b) Equal to that of $Cl^-$, that is 8

(c) Not equal to that of $Cl^-$, that is 6

(d) Not equal to that of $Cl^-$, that is 8

**33.** Which of the following statements is not true about NaCl structure

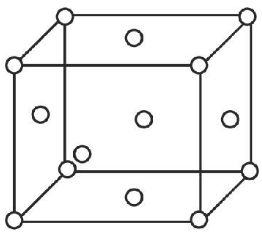
(a) $Cl^-$ ions are in fcc arrangement

(b) $Na^+$ ions has coordination number 4

(c) $Cl^-$ ions has coordination number 6

(d) Each unit cell contains 4NaCl molecules

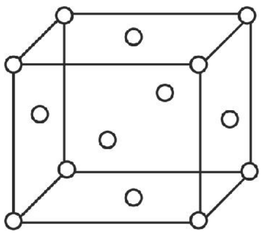
**34.** A solid $A^+B^-$ has the $B^-$ ions arranged as below. If the $A^+$ ions occupy half of the octahedral sites in the structure. The formula of solid is



(a) AB (b) $AB_2$

(c) $A_2B$ (d) $A_3B_4$

**35.** For the structure of solid given below if the lattice points represent $A^+$ ions & the $B^-$ ions occupy all the tetrahedral voids then coordination number of A is



(a) 2 (b) 4

(c) 6 (d) 8

**36.** In $A^+ B^-$ ionic compound, radii of $A^+$ and $B^-$ ions are 180 pm and 187 pm respectively. The crystal structure of this compound will be

(a) NaCl type (b) CsCl type

(c) ZnS type (d) Similar to diamond

**37.** If the value of ionic radius ratio $\left(\frac{r_c}{r_a}\right)$ is 0.52 in an ionic compound, the geometrical arrangement of ions in crystal is

(a) Tetrahedral (b) Planar

(c) Octahedral (d) Pyramidal

**38.** The maximum radius of sphere that can be fitted in the tetrahedral hole of cubical closed packing of sphere of radius r is.

(a) 0.732 r (b) 0.414 r

(c) 0.225 r (d) 0.155 r

**39.** In NaCl lattice the radius ratio is $\frac{r_{Na^+}}{r_{Cl^-}} =$

(a) 0.225 (b) 0.115

(c) 0.5414 (d) 0.471

**40.** For an ionic crystal of the general formula AX and coordination number 6, the value of radius ratio will be

(a) Greater than 0.73

(b) In between 0.73 and 0.41

(c) In between 0.41 and 0.22

(d) Less than 0.22

**41.** A unit cell of sodium chloride has four formula units. The edge length of unit cell is 0.564 nm. What is the density of sodium chloride?

(a) 3.89 g $cm^{-3}$ (b) 2.16 g $cm^{-3}$

(c) 3 g $cm^{-3}$ (d) 1.82 g $cm^{-3}$

## defects

**42.** Which defect causes decrease in the density of crystal

(a) Frenkel (b) Schottky

(c) Interstitial (d) F-centre

**43.** In a solid lattice the cation has left a lattice site and is located at an interstitial position, the lattice defect is

(a) Interstitial defect (b) Valency defect

(c) Frenkel defect (d) Schottky defect

**44.** The correct statement regarding F-centre is

(a) Electron are held in the voids of crystals

(b) F-centre produces colour to the crystals

(c) Conductivity of the crystal increases due to F- centre

(d) All of these

**45.** Schottky defect in crystals is observed when

(a) Density of crystal is increased

(b) Unequal number of cations and anions are missing from the lattice

(c) An ion leaves its normal site and occupies an interstitial site

(d) Equal number of cations and anions are missing from the lattice

**46.** If a electron is present in place of anion in a crystal lattice, then it is called

(a) Frenkel defect (b) Schottky defect

(c) Interstitial defect (d) F-centre

**47.** Schottky defect in crystals is observed when

(a) an ion leaves its normal site and occupies an interstitial site

(b) few cations are missing from the crystal

(c) equal number of cations and anions are missing from the crystal

(d) few anions are missing from the crystal.

**48.** What is the effect of Frenkel defect on the density of ionic solids?

(a) The density of the crystal increases.

(b) The density of the crystal decreases.

(c) The denisty of the crystal remains unchanged.

(d) There is no relationship between density of a crystal and defect present in it.

**49.** Alkali halides do not show Frenkel defect because

(a) cations and anions have almost equal size

(b) there is a large difference in size of cations and anions.

(c) cations and anions have low coordination number

(d) anions cannot be accommodated in voids.

**50.** Zinc oxide loses oxygen on heating according to the reaction,

$$ZnO \xrightarrow{heat} Zn^{2+} + \frac{1}{2}O_2 + 2e^-$$

It becomes yellow on heating because

(a) $Zn^{2+}$ ions and electrons move to interstitial sites and F-centres are created

(b) Oxygen and electrons move out of the crystal and ions become yellow

(c) $Zn^{2+}$ again combine with oxygen to give yellow oxide

(d) $Zn^{2+}$ are replaced by oxygen.

**51.** In a Schottky defect,

(a) an ion moves to interstitial position between the lattice points

(b) electrons are trapped in a lattice site

(c) some lattice sites are vacant

(d) some extra cations are present in interstitial spaces.

## Magnetic and Electrical Properties

**52.** Which substance shows antiferromagnetism?

(a) $ZrO_2$ (b) CdO

(c) $CrO_2$ (d) $Mn_2O_3$

**53.** The electricity produced on applying stress on the crystals is called

(a) Pyroelectricity (b) Piezoelectricity

(c) Ferroelectricity (d) Anti-ferroelectricity

**54.** Crystals where dipoles may align themselves in an ordered manner so that there is a net dipole moment, exhibit

(a) Pyro-electricity (b) Piezo-electricity

(c) Ferro-electricity (d) Antiferro-electricity

**55.** On heating some polar crystals, weak electric current is produced. It is termed as :

(a) piezoelectricity (b) pyroelectricity

(c) photoelectric current (d) superconductivity

**56.** Which one among the following is an example of ferroelectric substance?

(a) Quartz (b) lead chromate

(c) Barium titanate (d) Rochelle salt.

**57.** The substance which possesses zero resistance at OK is called

(a) Conductor (b) Superconductor

(c) Insulator (d) Semiconductor.

**58.** Which substance acts as superconductor at 4 K?

(a) Hg (c) Cu

(c) Na (d) Mg

**59.** A solid with high electrical and thermal conductivity from the following is

(a) Si (b) Li

(c) NaCl (d) Ice

**60.** Pure silicon and germanimum behave as

(a) conductors (b) semiconductors

(c) insulators (d) piezoelectric crystals.

# EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTION

**1.** Number of atoms in the unit cell of Na (bcc type crystal) and Mg (fcc type crystal) are, respectively **(2002)**

(a) 4, 4 (b) 4, 2

(c) 2, 4 (d) 1, 1

2. How many unit cells are present in a cube shaped ideal crystal of NaCl of mass 1.00 g ?

(Atomic mass : Na = 23, Cl = 35.5) **(2003)**

(a) $2.57 \times 10^{21}$ (b) $5.14 \times 10^{21}$

(c) $1.28 \times 10^{21}$ (d) $1.71 \times 10^{21}$

**3.** Glass is **(2003)**

(a) Microcrystalline solid (b) Super cooled liquid

(c) Gel (d) Polymeric mixture

**4.** What type of crystal defect is indicated in the diagram shown below ? **(2004)**

$Na^+ Cl^- Na^+ Cl^- Na^+ Cl^-$

$Cl^- Cl^- Na^+ Na^+$

$Na^+ Cl^- Cl^-, Na^+ Cl^-$

$Cl^- Na^+ Cl^- Na^+ Na^+$

(a) Frenkel defect (b) Schottky defect

(c) Interstitial defect

(d) Frenkel and Schottky defects

**5.** An ionic compound has a unit cell consisting of A ions at the corners of a cube and B ions on the centres of the faces of the cube. The empirical formula for this compounds would be **(2005)**

(a) $A_3B$ (b) $AB_3$

(c) $A_2B$ (d) AB

**6.** Total volume of atoms present in a face-centred cubic unit cell of a metal is (r is atomic radius) **(2006)**

(a) $\frac{20}{3}\pi r^3$ (b) $\frac{24}{3}\pi r^3$

(c) $\frac{12}{3}\pi r^3$ (d) $\frac{16}{3}\pi r^3$

**7.** The total volume of atoms present in face-centred cubic unit cell of a metal is (r is atomic radius) **(2008)**

(a) $\frac{20\pi r^3}{3}$ (b) $\frac{24\pi r^3}{3}$

(c) $\frac{12\pi r^3}{3}$ (d) $\frac{16\pi r^3}{3}$

**8.** In a compound, atoms of element Y form ccp lattice and those of element X occupy 2/3rd of tetrahedral voids. The formula of the compound will be **(2008)**

(a) $X_4Y_3$ (b) $X_2Y_3$

(c) $X_2Y$ (d) $X_3Y_4$

**9.** Copper crystallizes in fcc with a unit cell length of 361 pm. What is the radius of copper atom ? **(2007)**

(a) 109 pm (b) 127 pm

(c) 157 pm (d) 181 pm

**10.** Copper crystallises in fcc with a unit cell length of 361 pm. What is the radius of copper atom ? **(2009)**

(a) 108 pm (b) 127 pm

(c) 157 pm (d) 181 pm

**11.** Percentage of free space in cubic close packed structure and in body centred packed structure are respectively **(2010)**

(a) 30 % and 26% (b) 26% and 32%

(c) 32% and 48% (d) 48% and 26%

**12.** The edge length of a face centered cubic cell of an ionic substance is 508 pm. If the radius of the cation is 110 pm, the radius of the anion is **(2010)**

(a) 288 pm (b) 398 pm

(c) 618 pm (d) 144 pm

**13.** Copper crystallises in fcc lattice with a unit cell edge of 361 pm. The radius of copper atom is **(2011)**

(a) 181 pm (b) 108 pm

(c) 128 pm (d) 157 pm

**14.** In a face centred cubic lattice, atom a occupies the corner positions and atom B occupies the face centre positions. If one atom of B is missing from one of the face centred points, the formula of the compound is **(2011)**

(a) $A_2B$ (b) $AB_2$

(c) $A_2B_2$ (d) $A_2B_5$

**15.** Lithium forms body-centred cubic structure. The length of the side of its unit cell is 351 pm. Atomic radius of the lithium will be **(2012)**

(a) 75 pm (b) 300 pm

(c) 240 pm (d) 152 pm

**16.** Experimentally it was found that a metal oxide has formula $M_{0.98}O$. Mental M, present as $M^{2+}$ and $M^{3+}$ in its oxide. Fraction of the metal which exists as $M^{3+}$ would be

**(2013)**

(a) 7.01 % (b) 4.08 %

(c) 6.05 % (d) 5.08 %

**17.** CsCl crystallises in body centred cubic lattice. If ‘a’ is its edge length then which of the following expressions is correct ? **(2014)**

(a) $r_{Cs^+} + r_{Cl^-} = \frac{3a}{2}$ (b) $r_{Cs^+} + r_{Cl^-} = \frac{\sqrt{3}}{2}a$

(c) $r_{Cs^+} + r_{Cl^-} = \sqrt{3}a$ (d) $r_{Cs^+} + r_{Cl^-} = 3a$

**18.** Sodium metal crystallizes in a body centred cubic lattice with a unit cell edge of 4.29 Å. The radius of sodium atom is approximately: **(2015)**

(a) 5.72 Å (b) 0.93 Å

(c) 1.86 Å (d) 3.22 Å

**19.** A metal crystallises in a face centred cubic structure. If the edge length of its unit cell is ‘a’, the closest approach between two atoms in metallic crystal will be : **(2017)**

(a) $2\sqrt{2}a$ (b) $\sqrt{2}a$

(c) $\frac{a}{\sqrt{2}}$ (d) 2a

**20.** Which type of 'defect' has the presence of cations in the interstitial sites ? **(2018)**

(a) Frenkel defect (b) Metal deficiency defect

(c) Schottky defect (d) Vacancy defect

## JEE MAINS ONLINE QUESTION

**1.** In a face centered cubic lattice atoms A are at the corner points and atoms B at the face centered points. If atom B is missing from one of the face centered points, the formula of the ionic compound is: **Online 2014 SET (1)**

(a) $AB_2$ (b) $A_5B_2$

(c) $A_2B_3$ (d) $A_2B_5$

**2.** The appearance of colour in solid alkali metal halides is generally due to: **Online 2014 SET (2)**

(a) Schottky defect

(b) Frenkel defect

(c) Interstitial position

(d) F-centres

**3.** In a monoclinic unit cell, the relation of sides and angles are respectively: **Online 2014 SET (3)**

(a) a ‘‘‘ b ‘‘‘ c and α ‘‘‘ β ‘‘‘ γ ‘‘‘ $90^0$

(b) a ‘‘‘ b ‘‘‘ c and $\alpha = \beta = \gamma = 90^0$

(c) a ‘‘‘ b ‘‘‘ c and $\beta = \gamma = 90^0$ ‘‘‘ α

(d) a = b ‘‘‘ c and $\alpha = \beta = \gamma = 90^0$

**4.** The total number of octahedral void(s) per atom present in a cubic close packed structure is :

**Online 2014 SET (4)**

(a) 1 (b) 2

(c) 4 (d) 3

**5.** Which of the following exists as covalent crystals in the solid state? **Online 2015 SET (1)**

(a) Silicon (b) Sulphur

(c) Phosphorous (d) Iodine

**6.** An element (atomic mass 100g/mol) having bcc structure has until cells edge 400 pm. Then density of the element is : **Online 2016 SET (1)**

(a) 10.376 g/cm$^3$ (b) 5.188 g/cm$^3$

(c) 7.289 g/cm$^3$ (d) 2.144 g/cm$^3$

**7.** Which of the following arrangements shows the schematic alignment of magnetic moments of antiferromagnetic substance ? **Online 2018 SET (1)**

(1) $\uparrow \uparrow \uparrow \uparrow \uparrow \uparrow$

(2) $\uparrow \downarrow \downarrow \downarrow \downarrow \uparrow$

(3) $\uparrow \downarrow \uparrow \downarrow \uparrow \downarrow$

(4) $\uparrow \uparrow \downarrow \uparrow \uparrow \downarrow$

**8.** All of the following share the same crystal structure except : **Online 2018 SET (2)**

(1) LiCl (2) NaCl

(3) RbCl (4) CsCl

# EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

1. All questions marked **"S"** are single choice questions
2. All questions marked **"M"** are multiple choice questions
3. All questions marked **"C"** are comprehension based questions
4. All questions marked **"A"** are assertion–reason type questions
   **(A)** If both assertion and reason are correct and reason is the correct explanation of assertion.
   **(B)** If both assertion and reason are true but reason is not the correct explanation of assertion.
   **(C)** If assertion is true but reason is false.
   **(D)** If reason is true but assertion is false.
5. All questions marked **"X"** are matrix–match type questions
6. All questions marked **"I"** are integer type questions

## Characterstic Properties of Solid

**1. (S)** Which is not a property of solids
(a) Solids are always crystalline in nature
(b) Solids have high density and low compressibility
(c) The diffusion of solids is very slow
(d) Solids have definite volume

**2. (S)** Which of the following is a pseudo solid
(a) $CaF_2$ (b) Glass
(c) $NaCl$ (d) All of these

**3. (S)** Which of the following is non-crystalline solid
(a) CsCl (b) $NaCl$
(c) $CaF_2$ (d) Glass

**4. (S)** Among solids the highest melting point is established by
(a) Covalent solids (b) Ionic solids
(c) Pseudo solids (d) Molecular solids

**5. (S)** Particles of quartz are packed by
(a) Electrical attraction forces
(b) Vander Waal's forces
(c) Covalent bond forces
(d) Strong electrostatic attraction forces

**6. (S)** Which of the following is an example of covalent crystal solid
(a) Si (b) NaF
(c) Al (d) Ar

**7. (S)** Which of the following is an example of metallic crystal solid
(a) C (b) Si
(c) W (d) $AgCl$

**8. (S)** Which of the following is an example of ionic crystal solid
(a) Diamond (b) LiF
(c) Li (d) Silicon

**9. (S)** NaCl is an example of
(a) Covalent solid (b) Ionic solid
(c) Molecular solid (d) Metallic solid

**10. (S)** Diamond is an example of
(a) Solid with hydrogen bonding
(b) Electrovalent solid
(c) Covalent solid (d) Glass

**11. (M)** Graphite is
(a) a good conductor (b) $sp^3$ hybridized
(c) an amorphous solid (d) soft and slippery

**12. (M)** Diamond is
(a) a covalent solid (b) non-conductor
(c) soft and slippery (d) $sp^2$ hybridized

**13. (A)** **Assertion :** Crystalline solids are anisotropic.
**Reason :** The constituent particles are very closely packed.
(a) A (b) B
(c) C (d) D

14. (A) **Assertion :** Amorphous solids are isotropic.

**Reason :** Amorphous solids do not possess rigidity.

(a) A (b) B

(c) C (d) D

15. (A) **Assertion :** Ionic crystals have the highest melting point.

**Reason :** Covalent bonds are stronger than ionic bonds.

(a) A (b) B

(c) C (d) D

16. (X) Match the types of solid with their examples/properties.

| Column I | Column II |
|---|---|
| (A) Molecular solid | (p) Dry ice |
| (B) Covalent solid | (q) Coppe |
| (C) Metallic solid | (r) Generally behave as insulators |
| (D) Ionic solid | (s) Generally have low melting points |

## Crystal Lattice and Unit Cell

17. (S) Which of the following does not represent a type of crystal system

(a) Triclinic (b) Monoclinic

(c) Rhombohedral (d) Isotropical

18. (S) Example of unit cell with crystallographic dimensions $a \neq b \neq c, \alpha = \gamma = 90^\circ, \beta \neq 90^\circ$ is

(a) Calcite (b) Graphite

(c) Rhombic sulphur (d) Monoclinic sulphur

19. (S) Crystals can be classified into ...............basic crystal habits

(a) 3 (b) 7

(c) 14 (d) 4

20. (S) Tetragonal crystal system has the following unit cell dimensions

(a) $a = b = c$ and $\alpha = \beta = \gamma = 90^\circ$

(b) $a = b \neq c$ and $\alpha = \beta = \gamma = 90^\circ$

(c) $a \neq b \neq c$ and $\alpha = \beta = \gamma = 90^\circ$

(d) $a = b \neq c$ and $\alpha = \beta = 90^\circ, \gamma = 120^\circ$

21. (S) Monoclinic crystal has dimensions

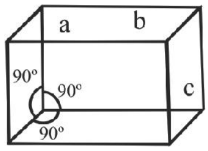
(a) $a \neq b \neq c, \alpha = \gamma = 90^\circ \beta \neq 90^\circ$

(b) $a = b = c, \alpha = \beta = \gamma = 90^\circ$

(c) $a = b \neq c, \alpha = \beta = \gamma = 90^\circ$

(d) $a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90^\circ$

22. (S) The unit cell with the structure below refer to ............crystal system




(a) Cubic (b) Orthorhombic

(c) Tetragonal (d) Trigonal

23. (S) What is the coordination number of sodium in $Na_2O$

(a) 6 (b) 4

(c) 8 (d) 2

24. (S) If an atom is present in the centre of the cube, the participation of that atom per unit cell is

(a) $\frac{1}{4}$ (b) 1

(c) $\frac{1}{2}$ (d) $\frac{1}{8}$

25. (M) Crystal systems in which no two axial lengths are equal are

(a) triclinic (b) orthorhombic

(c) monoclinic (d) tetragonal

26. (M) Which of the following systems do not give correct description of axial lengths and axial angles ?

(a) Hexagonal : $a = b \neq c, \alpha = \beta = 90^\circ, \gamma = 120^\circ$

(b) Trigonal : $a = b \neq c, \alpha = \beta = 90^\circ, \gamma = 90^\circ$

(c) Monoclinic : $a \neq b \neq c, \alpha = \beta = \gamma \neq 90^\circ$

(d) Cubic : $a = b \neq c, \alpha = \beta = \gamma = 90^\circ$

27. (A) **Assertion :** Triclinic system is the most unsymmetrical system.

**Reason :** All axial lengths are different in a triclinic system.

(a) A (b) B

(c) C (d) D

**28. (A)** **Assertion :** Graphite is an example of tetragonal crystal system.

**Reason :** For a tetragonal system, $a = b \neq c$, $\alpha = \beta = \gamma = 90°$.

(a) A (b) B
(c) C (d) D

## Comprehensiton

Density of a unit cell is the same as the density of the substances. So, if the density of the substance is known, we can calculate the number of atoms or dimesions of the unit cell. The density of the unit cell is related to its mass (M), number of atoms per unit cell (Z), edge length (a in cm), and Avogadro's constant $N_A$ as :

$$\rho = \frac{Z \times M}{a^3 \times N_A}$$

**29. (C)** An element X crystallizes in a structure having an fcc unit cell of an edge 100 pm. If 24 g of the element contains $24 \times 10^{23}$ atoms, the density is

(a) 2.40 g $cm^{-3}$ (b) 40 g $cm^{-3}$
(c) 4 g $cm^{-3}$ (d) 24 g $cm^{-3}$

**30. (C)** The number of atoms present in 100 g of a bcc crystal (density = 12.5 g $cm^{-3}$) having cell edge 200 pm is

(a) $1 \times 10^{25}$ (b) $1 \times 10^{24}$
(c) $2 \times 10^{24}$ (d) $2 \times 10^{26}$

**31. (C)** A metal A (atomic mass = 60) has a body - centered cubic crystal structure. The density of the metal is 4.2 g $cm^{-3}$. The volume of unit cell is

(a) $8.2 \times 10^{-23}$ $cm^3$ (b) $4.72 \times 10^{-23}$ $cm^3$
(c) $3.86 \times 10^{-23}$ $cm^3$ (d) None of these

**32. (I)** A metal X crystallizes in a unit cell in which the radius of atoms (r) is related to edge of unit cell (a) as r = 0.3535a. The total number of atoms present per unit cell is ______.

**33. (I)** A cubic unit cell has one atom on each corner and one atom on each body diagonal. The number of atoms in the unit cell is ______ .

**34. (X)** Match the type of packing with the metal possessing it/space occupied.

| Column I | Column II |
|---|---|
| (A) Body-centered cubic (bcc) | (p) Iron |
| (B) Hexagonal cubic | (q) 52% |
| (C) Cubic close packing | (r) 68% |
| (D) Simple cubic | (s) 74% |

## Density

**35. (S)** The formula for determination of density of unit cell is

(a) $\frac{a^3 \times N_0}{N \times M}$ g $cm^{-3}$ (b) $\frac{N \times M}{a^3 \times N_0}$ g $cm^{-3}$

(c) $\frac{a^3 \times M}{N \times N_0}$ g $cm^{-3}$ (d) $\frac{M \times N_0}{a^3 \times N}$ g $cm^{-3}$

**36. (S)** Potassium has a bcc structure with nearest neighbour distance 4.52 Å. Its atomic weight is 39. Its density (in kg $m^{-3}$) will be

(a) 454 (b) 804
(c) 852 (d) 911

**37. (S)** A metallic element has a cubic lattice. Each edge of the unit cell is 2 Å. The density of the metal is 2.5 g $cm^{-3}$ . The unit cells in 200 g of the metal are

(a) $1 \times 10^{25}$ (b) $1 \times 10^{24}$
(c) $1 \times 10^{22}$ (d) $1 \times 10^{20}$

**38. (S)** The density of KBr is 2.75 g $cm^{-3}$. The length of the unit cell is 654 pm. Atomic mass of K = 39, Br = 80. Then the solid is

(a) Face centred cubic
(b) Simple cubic system
(c) Body centred cubic system
(d) None of these

**39. (S)** A metal has bcc structure and the edge length of its unit cell is 3.04Å. The volume of the unit cell in $cm^3$ will be

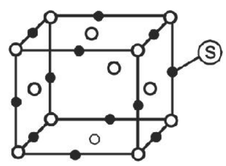
(a) $1.6 \times 10^{-21}$ $cm^3$ (b) $2.81 \times 10^{-23}$ $cm^3$
(c) $6.02 \times 10^{-23}$ $cm^3$ (d) $6.6 \times 10^{-24}$ $cm^3$

**40. (S)** A face-centred cubic element (atomic mass 60) has a cell edge of 400 pm. What is its density?

(a) 6.2 g $cm^{-3}$ (b) 24.8 g $cm^{-3}$
(c) 12.4 g $cm^{-3}$ (d) 3.1 g $cm^{-3}$

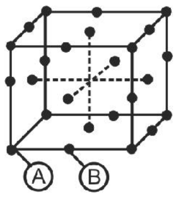
## Packing and Voids

**41. (S)** For the structure given below the site marked as S is a




(a) Tetrahedral void (b) Cubic void
(c) Octahedral void (d) None of these

**42. (S)** For a solid with the following structure, the co-ordination number of the point B is




(a) 3 (b) 4
(c) 5 (d) 6

**43. (S)** The coordination number of a cation occupying a tetrahedral hole is

(a) 6 (b) 8
(c) 12 (d) 4

**44. (S)** In hcp arrangement, the number of nearest neighbours are

(a) 8 (b) 6
(c) 4 (d) 12

**45. (S)** The coordination number of a metal crystallizing in a hexagonal close packed structure is

(a) 4 (b) 12
(c) 8 (d) 6

**46. (S)** The ratio of close-packed atoms to tetrahedral holes in cubic close packing is

(a) 1 : 1 (b) 1 : 2
(c) 1 : 3 (d) 2 : 1

**47. (S)** The number of octahedral voids in a unit cell of a cubical closest packed structure is

(a) 1 (b) 2
(c) 4 (d) 8

**48. (S)** In octahedral holes (voids)

(a) A simple triangular void surrounded by four spheres

(b) A bi-triangular void surrounded by four spheres

(c) A bi-triangular void surrounded by six spheres

(d) A bi-triangular void surrounded by eight spheres

**49. (S)** The number of tetrahedral voids in a unit cell of cubical closest packed structure is

(a) 1 (b) 2
(c) 4 (d) 8

**50. (M)** Which of the following are not true about hexagonal close packing ?

(a) It has a coordination number of 6.

(b) It has 26% empty space.

(c) It has ABAB..... type of arrangement

(d) It is as closely packed as body-centered cubic packing.

**51. (A)** **Assertion :** A hexagonal close packed structure is more closely packed than cubic-close packed structure.

**Reason :** Hexagonal close packing and cubic packing have coordination number of 12.

(a) A (b) B
(c) C (d) D

**52. (A)** **Assertion :** The number of tetrahedral voids is double the number of octahedral voids.

**Reason :** The size of the tetrahedral void is half of that of the octahedral void.

(a) A (b) B
(c) C (d) D

**53. (A)** **Assertion :** Size of tetrahedral void is much larger than an octahedral void.

**Reason :** The cations may occupy more space than anions in crystal packing.

(a) A (b) B
(c) C (d) D

## COMPREHENSION

X-ray studies show that the packing of atoms in a crystal of a metal is found to be in layer such that starting from any layer, every fourth layer is exactly identical. The density of the metal is found to be 19.4 g $cm^{-3}$ and its atomic mass is 197 u.

**54. (C)** The coordination number of metal atom in the crystal is

(a) 4 (b) 6
(c) 8 (d) 12

**55. (C)** The fraction occupied by metal atoms in the crystal is

(a) 0.52 (b) 0.68
(c) 0.74 (d) 1.0

**56. (C)** The approximate number of unit cells present in 1 g of metal is

(a) $3.06 \times 10^{21}$ (b) $1.53 \times 10^{21}$

(c) $3.82 \times 10^{20}$ (d) $7.64 \times 10^{20}$

**57. (C)** The length of the edge of the unit cell will be

(a) 407 pm. (b) 189 pm.

(c) 814 pm. (d) 204 pm.

**58. (C)** Assuming the metal atom to be spherical, its radius will be

(a) 103.5 pm. (b) 143.9 pm.

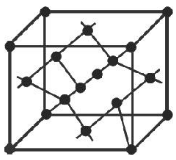
(c) 146.5 pm. (d) 267.8 pm.

**59. (I)** In hexagonal close packing, the difference in the number of tetrahedral and octahedral voids in a unit cell is ____.

**60. (I)** Atoms of element A form hcp arrangement and those of element B occupy 2/3rd of tetrahedral voids. The total number of A and B per formula unit is ______.

## Ionic Structures

**61. (S)** The following structure drawn is of



(a) Fluorite (b) Caesium chloride

(c) Wurtzite (d) Zinc blende

**62. (S)** The number of formula unit in unit cell of CsCl, type is

(a) 1 (b) 2

(c) 3 (d) 4

**63. (S)** The interionic distance for caesium chloride crystal will be

(a) a (b) $\frac{a}{2}$

(c) $\frac{\sqrt{3}a}{2}$ (d) $\frac{2a}{\sqrt{3}}$

**64. (S)** In CsCl lattice the coordination number of $Cl^-$ ion is

(a) 2 (b) 6

(c) 8 (d) 12

**65. (S)** If the coordination number of $Ca^{2+}$ in $CaF_2$ is 8, then the coordination number of $F^-$ ion would be

(a) 3 (b) 4

(c) 6 (d) 8

**66. (S)** How many chloride ions are there around sodium ion in sodium chloride crystal.

(a) 3 (b) 8

(c) 4 (d) 6

**67. (S)** The structure of MgO is similar to NaCl. What would be the coordination number of magnesium.

(a) 2 (b) 4

(c) 6 (d) 8

**68. (S)** In NaCl lattice the coordination number of $Cl^-$ ion is

(a) 2 (b) 4

(c) 6 (d) 8

**69. (S)** In the crystals, which of the following ionic compounds would you expect maximum distance between centres of cations and anions

(a) LiF (b) CsF

(c) CsI (d) LiI

**70. (S)** If the radius ratio is in the range of $0.732 - 1$, then the coordination number will be

(a) 2 (b) 4

(c) 6 (d) 8

**71. (S)** Which of the following contains zinc blende structure

(a) CuCl (b) CuBr

(c) AgI (d) All of these

**72. (S)** The number of unit cells in 58.5 g of NaCl is nearly

(a) $6 \times 10^{20}$ (b) $3 \times 10^{22}$

(c) $1.5 \times 10^{23}$ (d) $0.5 \times 10^{24}$

**73. (S)** Potassium fluoride has NaCl type structure. What is the distance between $K^+$ and $F^-$ ions if cell edge is a cm

(a) 2 a cm (b) a/2 cm

(c) 4a cm (d) a/4 cm

**74. (S)** The intermetallic compound LiAg crystallizes in cubic lattice in which both lithium and silver have coordination number of eight. The crystal class is

(a) Simple cube (b) Body-centred cube

(c) Face-centred cube (d) None of these

**75. (S)** Ferrous oxide has a cubic structure and each edge of the unit cell is 5.0 Å. Assuming density of the oxide as 4.0 g $cm^{-3}$, then the number of $Fe^{2+}$ and $O^{2-}$ ions present in each unit cell will be

(a) Four $Fe^{2+}$ and four $O^{2-}$ (b) Two $Fe^{2+}$ and four $O^{2-}$

(c) Four $Fe^{2+}$ and two $O^{2-}$ (d) Three $Fe^{2+}$ and three $O^{2-}$

**76. (S)** The ratio of cationic radius to anionic radius in an ionic crystal is greater than 0.732. Its coordination number is

(a) 6 (b) 8

(c) 1 (d) 4

**77. (S)** A solid $A^+ B^-$ has a body centred cubic structure. The distance of closest approach between the two ions is 0.767 Å. The edge length of the unit cell is

(a) $\frac{\sqrt{3}}{\sqrt{2}}$ pm (b) 142.2 pm

(c) $\sqrt{2}$ pm (d) 88.56 pm.

**78. (S)** A binary solid $x^+ y^-$ has a zinc blende structure with $y^-$ ions forming the lattice and $x^+$ ion occupying 25% tetrahedral sites. The formula of the solid is

(a) xy (b) $x_2y$

(c) $xy_2$ (d) $xy_4$

**79. (S)** In closest packing of A type of atom (radius $r_A$) the radius of atom B that can be fitted into octahedral void is

(a) $0.155\, r_A$ (b) $0.125\, r_A$

(c) $0.414\, r_A$ (d) $0.732\, r_A$

**80. (S)** A sample of electrically neutral NaCl crystal is analysed for its density which has some unoccupied sites. Two readings where taken.

1. Density of NaCl crystal assuming all sites are occupied = $2.178 \times 10^3$ kg m$^{-3}$

2. Density of NaCl crystal by not considering the unoccupied sites but only the occupied sites = $2.165 \times 10^3$ kg m$^{-3}$.

The percentage of unoccupied sites in NaCl crystal is

(a) $5.96 \times 10^{-2}$ (b) $5.96 \times 10^{-1}$

(c) 5.96 (d) 8.68

**81. (S)** $NH_4Cl$ crystallizes in a body centred cubic lattice, with a unit cell edge length of 387 pm. The distance between the oppositively charged ions in the lattice is :

(a) 335 pm (b) 154 pm

(c) 460 pm (d) 320 pm

**82. (M)** Which of the following statement are correct ?

(a) The coordination number each type of ion in CsCl crystal is 8.

(b) A metal that crystallizes in bcc structure has coordination number of 12.

(c) A unit cell of an ionic crystal shares some of its ions with other unit cells.

(d) The length of the edge of unit cell of NaCl is 552 pm

($r_{Na^+} = 95$ pm, $r_{Cl^-} = 181$ pm)

**83. (M)** The coordination number of anion is 4n

(a) sodium chloride (b) zinc chloride

(c) caesium chloride (d) calcium fluoride

**84. (M)** Which of the following statements are true ?

(a) An element with bcc structure has two atoms per unit cell.

(b) An ionic compound $A^+B^-$ with bcc structure has one AB formula unit per unit cell.

(c) The shape of the octahedral void is octahedral.

(d) The edge length of the crystal $A^+B^-$ is equal to the distance between $A^+$ and $B^-$ ions.

**85. (M)** In which of the following structures, the coordination number of both the ions is same ?

(a) Caesium chloride (b) Sodium chloride

(c) Zinc chloride (d) Sodium oxide

**86. (M)** Which of the following are true ?

(a) In NaCl crystals, $Na^+$ ions are present in all the octahedral voids.

(b) In ZnS (zinc blende), $Zn^{2+}$ ions are present in alternate tetrahedral voids.

(c) In $CaF_2$, $F^-$ ions occupy all the octahedral voids.

(d) In $Na_2O$, $O^{2-}$ ions occupy half of the octahedral voids.

**87. (M)** The coordination number of 8 for cation is found in

(a) CsCl (b) NaCl

(c) $CaF_2$ (d) $Na_2O$

**88. (M)** The density of KBr is 2.75 g $cm^{-3}$. The length of the unit cell is 654 pm. Atomic mass of K = 39, Br = 80. Then, which of the following statements are true ?

(a) It has 4 $K^+$ and 4 $Br^-$ ions per unit cell.

(b) It has a body-centered structure.

(c) It has rock-salt type structure.

(d) It can have Frenkel defects.

**89. (A)** **Assertion :** CsCl has face-centered cubic arrangement.

**Reason :** CsCl has one $Cs^+$ ion and one $Cl^-$ ions in its unit cell.

(a) A (b) B

(c) C (d) D

**90. (A)** **Assertion :** In $CaF_2$, $F^-$ ions occupy all the tetrahedral sites.

**Reason :** The number of $Ca^{2+}$ is double the number of $F^-$ ions.

(a) A (b) B

(c) C (d) D

**91. (A)** **Assertion :** Zinc blende and wurtzite both have fcc arrangement of sulphide ions.

**Reason :** There are four formula units of ZnS in both.

(a) A (b) B

(c) C (d) D

**92. (A)** **Assertion :** In NaCl crystal, all the octahedral voids are occupied by $Na^+$ ions.

**Reason :** The number of octahedral voids is equal to the number of $Na^+$ ions in the packing.

(a) A (b) B

(c) C (d) D

## Comprehensiton

Crystalline solids have an ordered internal arrangement of atoms due to which they exhibit symmetry. The atoms are arranged in a symmetrical fasion in the three-dimensional network known as space lattice. Each point in a crystal lattice is known as lattice point or lattice site.

Each point in a crystal lattice represents a constituent particle which can be an atom, ion or a molecule. The smallest repeating subunit in the lattice is known as unit cell. The unit cells are described as simple (points at all the corners), body-centered (points at all the corners and in the center), face-centered (points at all the corners and center of all faces), and end-centered (points at all the corners and centers of two opposite end faces) unit cells. For the stable ionic crystalline structures, the radius ratio limit for a cation to fit perfectly in the lattice of anions is determined by the radius ratio rule. This also defines the number of nearest neighbors of opposite charges surrounding the ion, which is known as the coordination number. This depends upon the ratio of radii of two types of ions, $r^+/r^-$. The radius ratios for coordination numbers 3, 4, 6 and 8 are 0.155 – 0.225–0.414, 0.414–0.732 and 0.732–1, respectively. The coordination number of ionic solids increases with increase in pressure and decreases with increase in temperature.

**93. (C)** The number of atoms per unit cell in primitive (sc), body-centered (bcc), face-centered (fcc) and end-centered (ecc) unit cell decreases as

(a) fcc > bcc > ecc > sc (b) fcc > bcc = ecc > sc

(c) bcc > fcc > sc = ecc (d) fcc > bcc > ecc = sc

**94. (C)** A metal crystallizes in a face-centered unti cell. Its edge length is 0.410 nm. The radius of the metal is

(a) 0.205 nm (b) 0.290 nm

(c) 0.145 nm (d) 0.578 nm

**95. (C)** In a cubic lattice of ABC, A atoms are present at all corners except one corner which is occupied by B atoms. C atoms are present at face centers. The formula of the compound is

(a) $A_8BC_{24}$ (b) $ABC_3$

(c) $A_7B_{24}C$ (d) $A_7BC_{24}$

**96. (C)** The ionic radii of $Na^+$, $K^+$ and $Br^-$ are 137, 148 and 195 pm, respectively. The coordinaiton number of cation in KBr and NaBr structures are, respectively,

(a) 8, 6 (b) 6, 4

(c) 6, 8 (d) 4, 6

**97. (I)** In NaCl structure, $Cl^-$ ions have ccp arrangement and $Na^+$ ions occupy all the octahedral sites. The toal number of $Na^+$ and $Cl^-$ ions per unit cell is ________.

**98. (I)** The radius ratio of an ionic solid $r_+/r_-$ is 0.524. The coordination number of this type of structure is ____.

**99. (I)** $NH_4^+$ and $Br^-$ ions have ionic radii of 139 pm and 186 pm, respectively. The coordination number of $NH_4^+$ ion in $NH_4Br$ is _____ .

## Defects

**100. (S)** Frenkel defect is caused due to

(a) An ion missing from the normal lattice site creating a vacancy

(b) An extra positive ion occupying an interstitial position in the lattice

(c) An extra negative ion occupying an interstitial position in the lattice

(d) The shift of a positive ion from its normal lattice site to an interstitial site

**101. (S)** Ionic solids, with Schottky defects, contain in their structure

(a) Equal number of cation and anion vacancies

(b) Anion vacancies and interstitial anions

(c) Cation vacancies only

(d) Cation vacancies and interstitial cations

**102. (S)** In Ag Br crystal, the ion size lies in the order $Ag^+ << Br^-$. The AgBr crystal should have the following characteristics

(a) Defectless (perfect) crystal

(b) Schottky defect only

(c) Frenkel defect only

(d) Both Schottky and Frenkel defects

**103. (M)** The density of KBr is 2.75 g $cm^{-3}$. The length of the unit cell is 654 pm. Atomic mass of K = 39, Br = 80. Then, which of the following statements are true ?

(a) It has 4 $K^+$ and 4 $Br^-$ ions per unit cell.

(b) It has a body-centered structure.

(c) It has rock-salt type structure.

(d) It can have Frenkel defects.

**104. (M)** In which of the following defects, the cations are present in interstitial sites ?

(a) Frenkel defect

(b) Schottky defect

(c) Metal deficient non-stoichometric compound

(d) Metal excess non-stoichiometric compound.

**105. (S)** If $Al^{3+}$ ions replace $Na^+$ ions at the edge centres of NaCl lattice, then the number of vacancies in 1 mole of NaCl will be

(a) $3.01 \times 10^{23}$ (b) $6.02 \times 10^{23}$

(c) $9.03 \times 10^{23}$ (d) $12.04 \times 10^{23}$

**106. (S)** Analysis of a sample of iron oxide shows that it has the formula $Fe_{0.90}O$. The fraction of iron present as $Fe^{2+}$ will be about

(a) 90% (b) 60%

(c) 78% (d) 70%

**107. (A)** **Assertion :** In ZnO, the excess $Zn^{2+}$ ions are present in interstitial sites.

**Reason :** Metal excess crystals have either missing cation or anion in interstitial site.

(a) A (b) B

(c) C (d) D

**108. (A)** **Assertion :** FeO is non-stoichiometric with $Fe_{0.95}O$.

**Reason :** Some $Fe^{2+}$ ions are replaced by $Fe^{3+}$ as $3Fe^{2+} = 2Fe^{3+}$ to maintain electric neutrality.

(a) A (b) B

(c) C (d) D

**109. (A)** **Assertion :** Frenkel defects are shown by silver halides.

**Reason :** The size of silver ions is large.

(a) A (b) B

(c) C (d) D

**110. (A)** **Assertion :** Due to Frenkel defect there is no effect on density of a solid.

**Reason :** Ions shift from lattice sites to interstitial sites in Frenkel defect.

(a) A (b) B

(c) C (d) D

**111. (I)** $Al^{3+}$ ions replace $Na^+$ ions at the edge centers of NaCl lattice. The number of vacancies in one mole Nacl is calculated as $n \times 10^{23}$. The value of n approximately is __.

**112. (X)** Match the imperfection in solids with the characteristic features.

| Column I | Column II |
|---|---|
| (A) Schottky defects | (p) Some cations occupy Interstitial sites |
| (B) Frenkel defects | (q) Conduct electricity due to free electrons |
| (C) Metal excess defects | (r) Act as p-type semiconductors |
| (D) Metal deficient defects | (s) Are non-stoichiometric defects |

## Magnetic and Eelctrical Properties

**113. (M)** Which of the following statements are not true ?

(a) The conductivity of metals increases with increase in temperature.

(b) Electrical conductivity of semiconductors increases with increase in temperature.

(c) A diode is a combination of conductor and a semiconductor.

(d) $V_2O_5$ behave as an insulator.

**114. (S)** An oxide of transition metal that has conductivity as well as appearance like that of copper is

(a) $Ti_2O_3$ (b) $V_2O_3$

(c) $ReO_3$ (d) $Mn_2O_3$

**115. (S)** The oxide that is insulator is

(a) VO (b) CoO

(c) $ReO_3$ (d) $Ti_2O_3$

**116. (S)** Silicon is a

(a) conductor (b) Semiconductor

(c) Nonconductor (d) Superconductor

**117. (S)** Which of the following is ferromagnetic?

(a) Ni (b) Co

(c) $Fe_3O_4$ (d) All are correct.

**118. (S)** Ferrimagnetic substances have

(a) zero magnetic moment

(b) small magnetic moment

(c) large magnetic moment

(d) any value of magnetic moment.

**119. (S)** Ferromagnetism is maximum in

(a) Fe (b) Ni

(c) Co (d) None

**120. (S)** Anti-ferromagnetic substances possess

(a) low magnetic moment

(b) large magnetic moment

(c) zero magnetic moment

(d) any value of magnetic moment

**121. (A)** **Assertion :** Antiferromagnetic substances become paramagnetic on heating to high temperature.

**Reason :** Heating results in spins of electrons becoming random.

(a) A (b) B

(c) C (d) D

## Comprehensiton

Crystals as such are never perfect and in general have some defect, which simply refers to a disruption in the periodic order of a crystalline material. Based on their nature, these are classified into three types : stoichiometric, impurity and non-stoichiometric defects. In non-stoichiometric defects, the ratio of the number of atoms of one kind to the number of atoms of the other kind does not correspond exactly to the ideal whole number ratio implied by the formula. For example, the ideal formula of ferrous oxide shoul dbe FeO, but in actually found to be $Fe_{0.93}O$ in a sample, because some ferric ions have replaced ferrous ions in the crystal. These defects affect the properties of the crystals. Sometimes such defects are introduced to improve the properties of the crystals for a specific use, such as in the field of electronics. Elements of Group 14 are commonly doped with impurities of elements of Group 13 or 15. In ionic crystals, impurities are introduced with cations having higher valence than the cations in the pure crystal structure.

**122. (C)** Which one of the following doping will produce p-type semi-conductor ?

(a) Silicon doped with arsenic

(b) Germanium doped with phosphorus

(c) Germanium doped with aluminium

(d) Silicon doped with phosphorus

**123. (C)** Which one of the following defects does not affect the density of the crystal ?

(a) Schottky defect (b) Interstitial defect

(c) Frenkel defect (d) Both b and c

**124. (C)** NaCl was doped with $10^{-3}$ mol % $SrCl_2$. The concentration of cation vacancies is

(a) $6.02 \times 10^{18}$ mol$^{-1}$ (b) $6.02 \times 10^{15}$ mol$^{-1}$

(c) $6.02 \times 10^{21}$ mol$^{-1}$ (d) $6.02 \times 10^{12}$ mol$^{-1}$

**125. (C)** The percentage of iron present as Fe(III) in $Fe_{0.93}O_{1.0}$ is

(a) 17.7% (b) 7.84%

(c) 11.5% (d) 9.6%

# EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTION

## Objective Questions - I (Only one correct option)

**1.** If the unit cell of a mineral has cubic close packed (ccp) array of oxygen atoms with m fraction of octahedral holes occupied by aluminium ions and n fraction of tetrahedral holes occupied by magnesium ions, m and n, respectively, are **(2015)**

(a) $\frac{1}{2}, \frac{1}{8}$ (b) $1, \frac{1}{4}$

(c) $\frac{1}{2}, \frac{1}{2}$ (d) $\frac{1}{4}, \frac{1}{8}$

**2.** Which of the following fcc structure contains cations in alternate tetrahedral voids ? **(2005)**

(a) NaCl (b) ZnS

(c) $Na_2O$ (d) $CaF_2$

**3.** A substance $A_xB_y$ crystallizes in a face centred cubic (fcc) lattice in which atoms 'A' occupy each corner of the cube and atoms 'B' occupy the centres of each face of the cube. Identify the correct composition of the substance $A_xB_y$ **(2002)**

(a) $AB_3$ (b) $A_4B_3$

(c) $A_3B$

(d) composition cannot be specified

**4.** In a solid 'AB' having the NaCl structure, 'A' atoms occupy the corners of the cubic unit cell. If all the face-centred atoms along one of the axes are removed, then the resultant stoichiometry of the solid is **(2001)**

(a) $AB_2$ (b) $A_2B$

(c) $A_4B_3$ (d) $A_3B_4$

**5.** The coordination number of a metal crystallizing in a hexagonal close-packed structure is **(1999)**

(a) 12 (b) 4

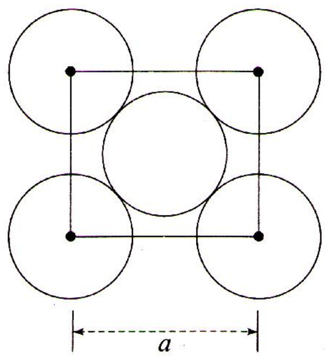
(c) 8 (d) 6

**6.** The bond energy (in kcal mol–1) of a C – C single bond is approximately **(1998)**

(a) 1 (b) 10

(c) 100 (d) 1000

**7.** The pakcing efficiency of the two-dimensional square unit cell shown below is **(1997)**




(a) 39.27% (b) 68.02%

(c) 74.05% (d) 78.54%

## Objective Questions II (One or more than one correct option)

**8.** The Correct statement(s) for cubic close packed (ccp) three dimensional structure is (are) **(2016)**

(a) The number of the nearest neighbours of an atom present in the topmost layer is 12

(b) The efficiency of atom packing is 74%

(c) The number of octahedral and tetrahedral voids per atom are 1 and 2, respectively

(d) The unit cell edge length is $2\sqrt{2}$ times the radius of the atom

**9.** The correct statement(s) regarding defects in solids is (are) **(1999)**

(a) Frenkel defect is usually favoured by a very small difference in the sizes of cation and anion.

(b) Frenkel defect is a dislocation defect

(c) Trapping of an electron in the lattice leads to the formation of F-center

(d) Schottky defects have no effect on the physical properties of solids.

**10.** Which of the following statments (s) is (are) correct ? **(1998)**

(a) The coordination number of each type of ion in CsCl crystal is 8

(b) A metal that crystallizes in bcc structure has a coordination number of 12

(c) A unit cell of an ionic crystal shares some of its ions with other unit cells

(d) The length of the unit cell in NaCl is 552 pm.

$(r_{Na^+} = 95 pm; r_{Cl^-} = 181 pm)$

**Comprehension (Ques. 11 to 13)**

In hexagonal systems of crystals, a frequently encountered arragenement of atoms is described as a hexagonal prism. Herem the top and bottom of the cell are regular hexagons and three atoms are sandwiched in between them. A space-filling model of this structure, called hexagonal close-packed (HCP), is constituted of a sphere on a flat surface surrounded in the same plane by six identical spheres as closely as possible. Three spheres are then placed over the first layer so that they touch each other and represent the second layer. Each one of these three spheres touches three spheres of the bottom layer. Finally, the second layer is covered with a third layer that is identical to the bottom layer that is identical to the bottom layer in relative position. Assume radius of every sphere to be 'r'. **(2008)**

**11.** The number of atoms on this HCP unit cell is :

(a) 4 (b) 6

(c) 12 (d) 17

**12.** The volume of this HCP unit cell is :

(a) $24\sqrt{2}r^3$ (b) $16\sqrt{2}r^3$

(c) $12\sqrt{2}r^3$ (d) $\frac{64r^3}{3\sqrt{3}}$

**13.** The empty space in this HCP unit cell is :

(a) 74% (b) 47.6%

(c) 32% (d) 26%

**Match the Column**

**14.** Match the crystal system/unit cells mentioned in Column-I with their characteristic features mentioned in Column-II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS. **(2007)**

| Column I | Column II |
|---|---|
| (A) Simple cubic and face-centred cubic | (p) have these cell parameters a = b = c and $\alpha = \beta = \gamma$ |
| (B) Cubic and rhombohedral | (q) are two crystal systems |
| (C) Cubic and tetragonal | (r) have only two crystallography angles of 90º |
| (D) Hexagonal and monoclinic | (s) belong to same crystal system |

**Assertion and Reason**

(A) If both ASSERTION and REASON are true and reason is the correct explanation of the assertion.

(B) If both ASSERTION and REASON are true but reason is not the correct explanation of the assertion.

(C) If ASSERTION is true but REASON is false.

(D) If ASSERTION is false but REASON is true.

**15.** **Assertion :** In any ionic soid (MX) with Schottky defects, the number of positive and negative ions are same.

**Reason :** Equal numbers of cation and anion vacancies are present. **(2001)**

(a) A (b) B

(c) C (d) D

## Subjective Questions

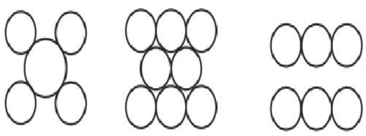
**16.** Sodium crystallizes in a bcc cubic lattice with the cell edge, a = 4.29 Å. What is the radius of sodium atom ? **(1994)**

**17.** A metallic element crystallizes into a lattice containing a sequence of layers of ABABAB...... Any packing of layers leaves out voids in the lattice. What percentage of this lattice is empty space ? **(1996)**

**18.** Chromium metal crystallizes with a body centred cubic lattice. The length of the unit edge is found to be 287 pm. Calculate the atomic radius. What would be the density of chromium in $g/cm^3$ ?

**19.** A metal crystllaizes into two cubic phases, face centred cubic (fcc) and body centred cubic (bcc), whose unit cell lengths are 3.5 and 3.0 A, respectively, Calculate the ratio of densities of fcc and bcc. **(1999)**

**20.** The figures given below show the location of atoms in three crystallographic planes in a fcc lattice. Draw the unit cell for the corresponding structures and identify these planes in your diagram.



**21.** (a) Marbles of diameter 10 mm are to be put in a square area of side 40 mm so that their centres are within this area.

(b) Find the maximum number of marbles per unit area and deduce an expression for calculating it. **(2003)**

**22.** The crsytal AB (rock salt structure) has molecular weight 6.023 y u. where, y is an arbitrary number in u. If the minimum distance between cation and anion is $y^{1/3}$ nm and the observed density is 20 $kg/m^3$. Find the (a) density in $kg/m^3$ and (b) type of defect **(2004)**

**23.** An element crystallizes in fcc lattice having edge length 400 pm. Calculate the maximum diameter of atom which can be placed in interstitial site without distorting the structure. **(2005)**

**24.** The edge length of unit cell of a metal having molecular weight 75 g/mol is 5 Å which crystallizes in cubic lattice. If the density is 2 g/cc then find the radius of metal atom. ($N_A = 6 \times 10^{23}$). Give the answer in pm. **(2006)**

**25.** A crystalline solid of a pure substance has a face-centred cubic structure with a cell edge of 400 pm. If the density of the substance in the crystal is 8 g $cm^{-3}$, then the number of atoms present in 256 g of the crystal is $N \times 10^{24}$. The value of N is **(2017)**

**26.** Consider an ionic solid MX with NaCl structure. Construct a new structure (Z) whose unit cell is constructed from the unit cell of MX following the sequential instructions given below. Neglect the charge balance.

(i) Remove all the anions (X) except the central one

(ii) Replace all the face centered cations (M) by anions (X)

(iii) Remove all the corner cations (M)

(iv) Replace the central anion (X) with cation (M)

The value of $\left(\dfrac{number\ of\ anions}{number\ of\ cations}\right)$ in Z is .......... .

**(2018)**

# ANSWER KEY

## EXERCISE - 1 : (Basic Objective Questions)

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1.** (d) | **2.** (c) | **3.** (d) | **4.** (a) | **5.** (d) | **6.** (a) | **7.** (d) | **8.** (d) | **9.** (a) | **10.** (b) |
| **11.** (b) | **12.** (c) | **13.** (c) | **14.** (c) | **15.** (a) | **16.** (d) | **17.** (a) | **18.** (d) | **19.** (b) | **20.** (b) |
| **21.** (b) | **22.** (b) | **23.** (a) | **24.** (b) | **25.** (c) | **26.** (a) | **27.** (d) | **28.** (d) | **29.** (b) | **30.** (d) |
| **31.** (c) | **32.** (b) | **33.** (b) | **34.** (b) | **35.** (d) | **36.** (b) | **37.** (c) | **38.** (b) | **39.** (c) | **40.** (b) |
| **41.** (b) | **42.** (b) | **43.** (c) | **44.** (d) | **45.** (d) | **46.** (d) | **47.** (c) | **48.** (c) | **49.** (a) | **50.** (a) |
| **51.** (c) | **52.** (d) | **53.** (b) | **54.** (b) | **55.** (b) | **56.** (c) | **57.** (b) | **58.** (a) | **59.** (b) | **60.** (b) |

## EXERCISE - 2 : (Previous Year JEE Mains Questions)

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1.** (c) | **2.** (a) | **3.** (b) | **4.** (b) | **5.** (b) | **6.** (d) | **7.** (d) | **8.** (a) | **9.** (b) | **10.** (b) |
| **11.** (b) | **12.** (d) | **13.** (c) | **14.** (d) | **15.** (d) | **16.** (b) | **17.** (b) | **18.** (c) | **19.** (c) | **20.** (a) |

### JEE Mains Online

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **1.** (d) | **2.** (d) | **3.** (c) | **4.** (b) | **5.** (a) | **6.** (b) | **7.** (c) | **8.** (d) |

## EXERCISE - 3 : (Advanced Objective Questions)

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1.** (a) | **2.** (b) | **3.** (d) | **4.** (a) | **5.** (c) | **6.** (a) | **7.** (c) | **8.** (b) | **9.** (b) | **10.** (c) |
| **11.** (ad) | **12.** (ab) | **13.** (b) | **14.** (c) | **15.** (d) | **16.** A – (p, r, s); B – ( r); C – (q); D – (r) | | | | **17.** (d) |
| **18.** (d) | **19.** (b) | **20.** (b) | **21.** (a) | **22.** (b) | **23.** (b) | **24.** (b) | **25.** (abc) | **26.** (bcd) | **27.** (b) |
| **28.** (d) | **29.** (b) | **30.** (c) | **31.** (b) | **32.** (0004) | **33.** (0005) | **34.** A – (r); B – (s); C – (p, s); D – (q) | | | |
| **35.** (b) | **36.** (d) | **37.** (a) | **38.** (a) | **39.** (b) | **40.** (a) | **41.** (c) | **42.** (d) | **43.** (d) | **44.** (d) |
| **45.** (b) | **46.** (b) | **47.** (c) | **48.** (c) | **49.** (d) | **50.** (a, d) | **51.** (d) | **52.** (c) | **53.** (d) | **54.** (d) |
| **55.** (c) | **56.** (d) | **57.** (a) | **58.** (b) | **59.** (0006) | **60.** (0007) | **61.** (d) | **62.** (a) | **63.** (c) | **64.** (c) |
| **65.** (b) | **66.** (d) | **67.** (c) | **68.** (c) | **69.** (c) | **70.** (d) | **71.** (d) | **72.** (c) | **73.** (b) | **74.** (b) |
| **75.** (a) | **76.** (b) | **77.** (d) | **78.** (c) | **79.** (c) | **80.** (b) | **81.** (a) | **82.** (acd) | **83.** (bd) | **84.** (ab) |
| **85.** (ab) | **86.** (ab) | **87.** (ac) | **88.** (ac) | **89.** (d) | **90.** (c) | **91.** (d) | **92.** (a) | **93.** (b) | **94.** (c) |
| **95.** (d) | **96.** (a) | **97.** (0008) | **98.** (0006) | **99.** (0008) | **100.** (d) | **101.** (a) | **102.** (c) | **103.** **(ac)** | **104.** (ad) |
| **105.** (a) | **106.** (c) | **107.** (c) | **108.** (a) | **109.** (c) | **110.** (a) | **111.** (0003) | | | |
| **112.** A – (r); B – (p) ; C – (p, q, s); D – (r, s) | | | | | **113.** (ac) | **114.** (c) | **115.** (b) | **116.** (b) | **117.** (d) |
| **118.** (b) | **119.** (a) | **120.** (c) | **121.** (a) | **122.** (c) | **123.** (c) | **124.** (a) | **125.** (c) | | |

## EXERCISE - 4 : (Previous Year JEE Advanced Questions)

**1.** (a) **2.** (b) **3.** (a) **4.** (d) **5.** (a) **6.** (c) **7.** (d) **8.** (b,c,d) **9.** (b,c) **10.** (a, c, d)

**11.** (b) **12.** (a) **13.** (a) **14.** $A \rightarrow p, s$ ; $B \rightarrow p, q$ ; $C \rightarrow q$ ; $D \rightarrow q, r$ **15.** (a) **16.** 1.86Å **17.** 26%

**18.** 124.3 pm, 7.3 **19.** 1.26 **21. (a)** 25, **(b)** $N = \left(\frac{x}{d} + 1\right)^2$ **22. (a)** 5Kg/m3, **(b)** metal excess defect

**23.** 117 pm **24.** 217 pm **25.** (2) **26.** (6)

***Dream on !!***